

AP Psychology — Complete Review

Prepared for **Full Knowledge Base**

September 27, 2026

Concepts covered: 79

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Unit 0 — Research Design (Research Methods)

Course Structure, Exam Format & Science Practices

id: res-course-exam-overview · Unit 0 — Research Design (Research Methods) · **core** exam science-practices
frq mcq

AP Psychology is the scientific study of behavior and mental processes, equivalent to an introductory college course with no prerequisites. All psychological phenomena have biological underpinnings — the course begins with neuroscience and builds toward complex social and clinical applications.

Exam structure:

- **Section I — Multiple choice:** 75 questions, 90 minutes, 66.7% of the score. About 65% concept application, 25% research methods and design, 10% data interpretation.
- **Section II — Free response:** 2 questions, 70 minutes, 33.3% of the score.
 - **AAQ (Article Analysis Question):** one summarized peer-reviewed study; 7 points; 25 minutes (10 for reading); assesses science practices 2, 3, and 4.
 - **EBQ (Evidence-Based Question):** three summarized sources on one topic; 7 points; 45 minutes (15 for reading); assesses practices 1 and 4.
- **Units (each 15-25% of the exam):** Unit 1 Biological Bases of Behavior; Unit 2 Cognition; Unit 3 Development and Learning; Unit 4 Social Psychology and Personality; Unit 5 Mental and Physical Health. Research design is assessed across all units through Skills 2 and 3.

The four science practices:

1. **Concept Application:** apply perspectives, theories, concepts, and research findings to scenarios; compare and contrast the seven perspectives (biological, behavioral, cognitive, evolutionary, humanistic, psychodynamic, sociocultural); explain how culture and cognitive biases apply.
2. **Research Methods and Design:** determine experimental vs. non-experimental methodology; evaluate design elements; evaluate ethics.
3. **Data Interpretation:** identify concepts in data; calculate and interpret mean, median, mode, range, standard deviation, and percentile rank; interpret skewness, bimodal distributions, regression toward the mean, correlation coefficients, effect sizes, and statistical significance.
4. **Argumentation:** propose defensible claims and provide reasoning grounded in scientific evidence to support, refute, or modify claims.

Exam tip: research design pays off in EVERY unit — IV/DV identification, sampling, and statistics appear inside scenario questions everywhere on the exam.

Historical Foundations: Wundt & the First Psychology Lab

id: res-wundt-history · Unit 0 — Research Design (Research Methods) · **core** history wundt foundations

Wilhelm Wundt — the “Father of Psychology”: established the **first psychology laboratory** in 1879 in Leipzig, Germany, marking psychology’s emergence as a separate scientific discipline rather than a branch of philosophy or physiology. Wundt and his students pioneered using controlled experiments to study mental processes and behavior.

Why it matters: psychology’s identity as a *science* rests on this move — replacing armchair speculation about the mind with systematic, measurable observation. Every research-methods concept in this unit descends from that step.

Context: earlier thinkers studied “the mind” philosophically; Wundt’s lab made the mind an object of experimental study. (The specific schools he inspired — structuralism and functionalism — and the introspection method are beyond the CED’s required terms.)

Misconception: Wundt did not “discover” psychology out of nothing, and he is not the father of *American* psychology — he founded the first *laboratory* and is called the father of experimental/scientific psychology.

Exam tip: a question stem naming “the first psychology laboratory” or “the father of psychology” points to Wundt; the expected fact is Leipzig, 1879.

Basic vs. Applied Research

id: res-basic-applied-research · Unit 0 — Research Design (Research Methods) · **core** research methods
basic-research applied-research

Basic research: pure science aimed at increasing the general knowledge base, with no immediate practical goal — e.g., studying how memory consolidation works, or how neurons fire.

Applied research: research conducted to solve specific, practical problems — e.g., designing a more effective study technique, testing a therapy, or reducing workplace accidents.

The relationship: applied work **BUILDS** on basic findings — basic research discovers that spaced practice beats cramming; applied research turns that into a classroom scheduling policy. The same question can be studied in either mode depending on the goal.

Misconception: “basic” does not mean simple, and “applied” is not automatically more valuable — basic research asks “what is true?”; applied research asks “what works, for whom, in this setting?”

Exam tip: classify a study by its **GOAL**: if the aim is general knowledge, it is basic; if the aim is fixing or improving something concrete, it is applied.

Experimental Method & Variables

id: res-experimental-method · Unit 0 — Research Design (Research Methods) · **core** experiment variables
causation methods

Experimental method: the **ONLY** method that can establish **cause-and-effect**. It requires (1) manipulation of an independent variable and (2) **random assignment** of participants to groups. If a study has random assignment, it is an experiment.

- **Independent variable (IV):** the variable the researcher *manipulates*; it influences the dependent variable. Ask: “What did they set up / vary?”
- **Dependent variable (DV):** the variable that is *measured*; it depends on the IV. Ask: “What did they measure?”
- **Confounding variable:** any difference between the experimental and control conditions *other than* the IV that might affect the DV.
- **Quasi-experiment:** looks like an experiment but lacks random assignment (e.g., comparing pre-existing groups); cannot support causal conclusions.
- **Field experiment:** conducted in the real world rather than the laboratory, trading control for realism.

Operational definition: a definition of a variable that is quantifiable — it states exactly how the variable was measured or administered so the study can be replicated.

- NOT operational: “happiness = participants’ reaction to the movie”
- Operational: “happiness = the number of times participants smiled during the movie”

Misconception: random **ASSIGNMENT** (who goes into which condition) is not random **SAMPLING** (who is drawn from the population) — assignment supports causation; sampling supports generalizability.

Exam tip: AAQ Part A wants ONE method name; AAQ Part B wants the operational definition of a NAMED variable, stated as a number or count. On EBQ claims, use causal language (“causes”, “improves”) only when the cited source was an experiment.

Non-Experimental Methods & Correlation

id: res-nonexperimental-methods · Unit 0 — Research Design (Research Methods) · **core** correlation methods
case-study observation meta-analysis

Non-experimental methods describe, relate, and observe — but cannot establish cause-and-effect.

1. **Case study:** an in-depth examination of one person or a small group, usually about an uncommon experience (Phineas Gage, H.M.). Cannot establish causation; limited generalizability.
2. **Naturalistic observation:** observing participants in their natural environment without interference or manipulation; high ecological validity, but no causal conclusions.
3. **Correlational study:** measuring the relationship between two or more variables without manipulation; often uses surveys or questionnaires. **Correlation does not equal causation.**
 - **Positive correlation:** both variables increase or decrease together.
 - **Negative correlation:** as one variable increases, the other decreases.
 - **Correlation coefficient (r):** ranges from -1 (perfect negative) to $+1$ (perfect positive); 0 = no relationship; the closer to plus or minus 1, the stronger. Data are displayed on **scatterplots**.
 - **Directionality problem:** we cannot know which variable influences the other.
 - **Third-variable problem:** an unmeasured variable may cause both (e.g., heat explains both ice-cream sales and drownings).
 - **Illusory correlation:** perceiving a relationship that does not exist.
 - **Correlational phrasing on exams:** “tend to,” “is related to,” “is associated with,” “has a relationship with,” “are likely to.” Causal language (“causes”) belongs **ONLY** to true experiments.
4. **Meta-analysis:** a statistical synthesis of the results of *multiple* studies; calculates an overall effect size across studies. The researcher does not run a new experiment.

Misconception: a strong negative correlation ($r = -0.9$) is a **VERY** strong relationship — the sign gives direction, the size gives strength. And a near-zero r does not prove “no relationship of any kind,” only no *linear* one.

Exam tip: AAQ Part A offers “correlational study” and “observational study” as method answers. If the prompt reports a measured relationship between two variables, choose correlational; if it reports watching behavior without manipulation, choose observational.

Sampling, Populations & Generalizability

id: res-sampling-generalizability · Unit 0 — Research Design (Research Methods) · **core** sampling generalizability
population methods

- **Population:** the entire group of interest from which a sample is drawn.
- **Sample:** the subset of participants actually studied.
- **Random sampling (random selection):** every member of the population has an equal chance of being selected — this affects *generalizability*.
- **Representative sample:** accurately reflects the population’s characteristics (gender ratio, age, location, education, income, ethnicity).
- **Convenience sampling:** selecting whoever is easily available (often the researcher’s own students); prone to bias.
- **Stratified sampling:** ensuring the sample represents the population on specific criteria (e.g., the same percentage of each class year).
- **Sampling bias:** when the sample systematically misrepresents the population.

- **Generalizability:** the extent to which results apply to the broader population. Evaluate by comparing sample demographics to the target population.

Response format for FRQs: answer with ONE side — “This study IS generalizable to X because the sample was randomly selected from X” OR “IS NOT generalizable because the sample does not represent X” (citing specific variables such as demographics, settings, or contexts). Do not simply say “the sample is too small” or “too diverse” without logic.

Misconception: random assignment and random sampling are different tools — random sampling (selection) governs who is IN the study and thus generalizability; random assignment governs which CONDITION participants land in and thus causation.

Exam tip: AAQ Part E asks for the extent of generalizability with SPECIFIC participant evidence — quote the sample description from the prompt (ages, setting, recruitment) and tie it to the target population.

Design Elements, Placebo, Blinding & Bias

id: res-design-controls-bias · Unit 0 — Research Design (Research Methods) · core design bias placebo blinding confounds

Groups and placebos:

- **Experimental group:** receives the treatment (the IV).
- **Control group:** does not receive the treatment; serves as the baseline for comparison.
- **Placebo:** an inert substance or sham treatment given so participants believe they are receiving the real one.
- **Placebo effect:** participants experience a change despite receiving no active substance.

Blinding:

- **Single-blind procedure:** participants do not know whether they are in the experimental or control group — controls for participant expectations.
- **Double-blind procedure:** neither participants *nor* the researchers interacting with them know the group assignments — controls for **experimenter bias** (the tendency for researchers to treat groups differently).

Sources of bias:

- **Participant bias / demand characteristics:** participants change behavior because they know they are being observed or hold expectations about the study’s purpose.
- **Social desirability effect (bias):** participants answer or behave in ways that make themselves look good rather than truthfully (overreporting healthy habits, underreporting prejudice) — a form of participant bias in self-report data; reduced by anonymity and confidentiality.
- **Experimenter bias:** researcher expectations influence results; the related concept **demand characteristics** is taught as a form the CED does not name explicitly.

Scientific process:

- **Hypothesis:** a testable prediction about the relationship between variables.
- **Falsifiability:** a hypothesis must be testable and disprovable.
- **Replication and peer review:** conclusions evolve through repeated, reviewed studies.

Misconception: the placebo effect is not “imagined illness” — real, measurable improvement occurs; that is exactly why control groups need placebos.

Exam tip: name the bias AND the fix as a pair: participant expectations — single-blind; experimenter expectations — double-blind; self-report image management — anonymity/confidentiality.

Descriptive Statistics

id: res-descriptive-statistics · Unit 0 — Research Design (Research Methods) · **core** statistics central-tendency variation distributions

Descriptive statistics: numerical summaries that organize and describe the main features of a data set — measures of central tendency (mean, median, mode) and measures of variation (range, variance, standard deviation). (Contrast: **inferential statistics** use sample data to draw conclusions about a population — e.g., testing for statistical significance.)

Measures of central tendency:

- **Mean (average):** sum of scores divided by the number of scores; skewed by outliers.
- **Median:** the midpoint of a distribution — the “true middle”; resistant to outliers.
- **Mode:** the most frequently occurring score.

Measures of variation:

- **Range:** highest score minus lowest score — how spread out the data are.
- **Variance:** dispersion from the mean; higher variance = more spread out.
- **Standard deviation (SD):** how tightly data cluster around the mean — small SD = close to the mean; large SD = more spread out.

Distributions:

- **Normal curve:** symmetrical bell curve; about 68% of scores fall within plus-or-minus 1 SD of the mean and about 95% within plus-or-minus 2 SD. Standardized IQ tests: mean = 100, SD = 15.
- **Percentile rank:** how much of the data falls BELOW a particular score.
- **Skewness:** the tail points toward the outliers. **Positively (right) skewed:** a few high scores pull the mean ABOVE the median. **Negatively (left) skewed:** a few low scores pull the mean BELOW the median.
- **Bimodal distribution:** two peaks (modes) in the data.
- **Regression toward the mean:** extreme scores tend to become less extreme (closer to average) with repeated measurement.

Misconception: in a skewed distribution the MEAN follows the tail, not the median — which is why the median is reported for skewed data like incomes.

Exam tip: AAQ Part C asks what a statistic *indicates about the study*, with direction — never a definition. Compare groups: “the meditation group’s mean of 6 was lower than the control group’s mean of 9, indicating less anxiety.”

Effect Size & Statistical Significance

id: res-effect-size-significance · Unit 0 — Research Design (Research Methods) · **standard** statistics effect-size significance frq

Effect size: how meaningful a difference or relationship is *in the real world* — practical significance.

- Large effect size: practical significance; strong correlation; large impact of the IV on the DV.
- Small effect size: limited practical applications; weak correlation; tiny impact.
- Common benchmarks: about 0.2 or less = small; 0.3-0.7 = medium; 0.8 or more = large.

Statistical significance: indicates results are *not likely due to chance* and are likely due to the IV.

- Goal: p at or below 0.05 — at most a 5% likelihood the results are due to chance, 95% likelihood they are due to the variables studied.
- p above 0.05 = NOT statistically significant — a greater chance the results are random.

FRQ response formats:

- **Effect size:** “The impact of the [IV] on the [DV] was strong/weak.” Use strength language for the impact, and always relate the statistic to the study WITH DIRECTION (“Group A scored higher than Group B”). Never just restate or define the statistic.
- **Significance:** explain the mean/SD meaning in context, then add “the results then showed that this was not likely due to chance.”

Misconception: statistical significance is NOT practical importance — a huge sample can make a trivial difference “significant,” which is why effect size is reported alongside p-values.

Exam tip: the two concepts answer different questions — significance asks “is it probably real (not chance)?”; effect size asks “is it big enough to matter?”

Cognitive Biases in Research & Thinking

id: res-cognitive-biases · Unit 0 — Research Design (Research Methods) · [standard](#) [bias](#) [reasoning](#) [research](#)

Cognitive biases are systematic errors in thinking that distort how we gather and evaluate evidence — in daily life AND in research:

- **Hindsight bias:** the tendency to perceive past events as more predictable than they were — the “knew-it-all-along” effect. After learning an outcome, people feel they “always knew” it.
- **Confirmation bias:** seeking, noticing, and remembering information that confirms existing beliefs while ignoring disconfirming evidence — e.g., reading only sources that agree with your view of a study.
- **Overconfidence:** overestimating the accuracy of one’s own beliefs and judgments — people are far more certain than they are correct.

How they threaten research: hindsight bias inflates confidence in post-hoc explanations; confirmation bias leads researchers (and participants) to notice supportive data only; overconfidence prevents the humility that replication demands. The remedies are methodological: pre-register hypotheses, seek disconfirming evidence, and replicate.

Misconception: these are errors of BELIEVING and REASONING, not memory failures per se — hindsight bias distorts judgment about predictability, even when the memory itself is accurate.

Exam tip: Concept Application items pair a scenario with the bias — “after the election, ‘I knew they would win’” = hindsight; “only bookmarking articles that agree with me” = confirmation; “I am 100% sure I can finish in an hour” (it takes three) = overconfidence.

Research Ethics & the IRB

id: res-ethics-irb · Unit 0 — Research Design (Research Methods) · [core](#) [ethics](#) [irb](#) [consent](#) [deception](#)

Institutional Review Board (IRB): the committee that reviews research proposals for human and non-human animal research BEFORE it begins.

- **Informed consent:** participants know and agree to what they are doing. **Informed assent** is the child version — children agree along with parental consent.
- **Protection from harm:** minimize physical and psychological risk; any risk must be justified.
- **Confidentiality and anonymity:** participants’ identities and data are protected. **Anonymity** = identity unknown even to researchers; **confidentiality** = identity known to researchers but protected.
- **Deception:** may only be used when necessary and justified; may involve **confederates** — actors posing as participants.
- **Debriefing:** after the study, participants are fully informed about its nature and purpose — essential whenever deception was used.
- **Voluntary participation:** participation must be voluntary; no coercion and no pressure to take part.

Misconception: deception is not banned — it is permitted when justified, PROVIDED participants are debriefed afterward. And a signed consent form does not excuse coercion; consent must be freely given.

Exam tip: AAQ Part D asks you to identify ONE ethical guideline that was EXPLICITLY used in the study and describe HOW it was applied. Only pick guidelines listed in the prompt — claiming an unaddressed guideline (or pairing it with a contradiction) costs the point.

AAQ: Article Analysis Question (FRQ 1)

id: res-aaq · Unit 0 — Research Design (Research Methods) · standard frq aaq skills statistics

AAQ (Article Analysis Question): FRQ 1 — one summarized peer-reviewed study, 7 points, 25 minutes (10 minutes reading time). Six parts:

- **Part A (1 pt):** identify the research method — case study, experiment, meta-analysis, observational study, or correlation. Answer with ONE method only; doubling up backfires, and mentioning surveys or cross-sectional designs as adjectives will not score.
- **Part B (1 pt):** state the operational definition of a NAMED variable — it must be quantifiable. Be specific, include scales if described, and focus on how the variable was measured or administered.
- **Part C (1 pt):** describe what a statistic (mean, median, mode, range, SD, percentile, correlation coefficient, effect size, statistical significance) *indicates about the study* — with direction. There is no point for restating or defining the statistic.
- **Part D (1 pt):** identify ONE ethical guideline explicitly used in the study AND describe how it was applied (IRB approval, informed consent/assent including the right to withdraw, protection from harm, confidentiality/anonymity, deception/confederates, debriefing). Only choose guidelines LISTED in the prompt — identifying unaddressed guidelines scores as contradiction, and any contradiction costs the point.
- **Part E (1 pt):** explain the extent to which the findings are generalizable, using specific participant evidence. Choose one side (or support both if claiming “partly”); reference specific variables (demographics, settings) or a relevant larger population. “Too diverse” without logic scores nothing.
- **Part F (2 pts):** explain how at least one research finding supports or refutes the hypothesis or concept. Step 1: identify the evidence. Step 2: explain HOW it supports or refutes — define the concept, then apply the results. Use causal language (“cause”) ONLY if the study was experimental in Part A.

Exam tip: time discipline matters — Part F is worth two points, so give it the most writing; the one-point parts need one precise sentence each.

EBQ: Evidence-Based Question (FRQ 2)

id: res-ebq · Unit 0 — Research Design (Research Methods) · standard frq ebq skills argumentation

EBQ (Evidence-Based Question): FRQ 2 — three summarized sources on one topic, 7 points, 45 minutes (15 minutes reading). Three parts:

- **Part A (1 pt):** propose a specific, defensible CLAIM — not a description; it must suggest an effect or recommendation (e.g., “sleep education improves adolescent mood,” not “the sources discuss sleep”).
- **Part B(i) (1 pt):** support the claim with one correctly CITED piece of specific evidence from a source — parenthetical “(Source 1)” or embedded “According to Source 1, ...”.
- **Part B(ii) (2 pts):** explain HOW the evidence supports the claim by applying a psychological perspective, theory, concept, or research finding.
- **Part C(i) (1 pt):** a DIFFERENT piece of cited evidence from a DIFFERENT source.
- **Part C(ii) (2 pts):** explain with a DIFFERENT psychological concept than the one used in B(ii).

Scoring traps: reusing the same concept for both explanations caps the explanation points; citing a source without specific evidence (just naming the study) is not support; the claim must be arguable, not a summary.

Exam tip: pick your two sources and two DIFFERENT concepts during the 15-minute reading window; write the claim first and keep every sentence tied to it.

Common Exam Errors & FRQ Conventions

id: res-common-pitfalls · Unit 0 — Research Design (Research Methods) · [standard](#) [exam](#) [frq](#) [pitfalls](#)

Common student errors documented from exam data:

1. **Confusing classical and operant conditioning:** classical = associating two *stimuli* to elicit an involuntary response; operant = *consequences* encouraging or discouraging voluntary behavior.
2. **Confusing conformity and obedience:** conformity = matching peers or implicit group norms; obedience = complying with explicit authority directives.
3. **Confusing OCD and OCPD:** the disorder (obsessions and compulsions) vs. the personality disorder (rigid perfectionism).
4. **Restating definitions instead of applying them:** FRQ points come from applying concepts to the scenario, never from textbook definitions alone.
5. **Using causal language in correlational studies** — this costs AAQ Part F points. “Causes” only for true experiments.
6. **Restating rather than interpreting statistics** — statistics must be described in relation to the study, with direction.
7. **Identifying brain structures when asked about functions** — the exam emphasizes functional connections to behavior.

Universal FRQ habits: answer the question ASKED, use the scenario’s names and variables, one concept per point line, and never contradict yourself — a single contradiction (e.g., “informed consent AND random assignment” when only one appears) can void the point.

Exam tip: before writing, underline the verb in the prompt — identify, describe, explain, and apply demand different depth, and “explain” requires a HOW, not a WHAT.

Unit 1 — Biological Bases of Behavior

Interaction of Heredity and Environment

id: bio-heredity-environment · Unit 1 — Biological Bases of Behavior · [core](#) [nature-nurture](#) [genetics](#) [evolution](#)
[predisposition](#)

Nature vs. nurture debate: the question of the relative importance of innate genetic qualities (**nature** — heredity; genetic, predisposed characteristics influencing physical, behavioral, and mental traits) versus personal experience (**nurture** — external factors such as family interactions or education). Modern psychology views them as *interacting*, not competing; every trait reflects both.

Genetic predisposition: an inherited likelihood of developing particular traits or conditions (e.g., a predisposition toward anxiety, depression, or height). A predisposition is a *tendency*, not a destiny — environment still shapes whether and how it appears.

Evolutionary perspective: explores how **natural selection** affects the expression of behavior and mental processes to increase survival and reproductive success. Behaviors that aided ancestors’ survival (fear of snakes, social bonding) may be inherited because they helped the species survive and reproduce.

Eugenics: a historical misapplication of evolutionary thinking — the discriminatory selection of “desired” heritable characteristics supposedly to “improve” future generations. It is cited as an ethical caution, not a legitimate science.

Misconception: “nature vs. nurture” is not an either/or choice. A student predisposed toward musical ability still needs exposure and practice; a supportive environment cannot grow a trait with no genetic footing. On FRQs, show the *interaction* (genes shape how we respond to environments; environments shape how genes are expressed).

Exam tip: when asked to apply the interaction of heredity and environment, name BOTH sides for the specific behavior in the scenario — e.g., “the twins share genes for impulsivity (nature), but the sibling raised in a chaotic home expresses it more (nurture).”

Behavior Genetics: Twin, Family & Adoption Studies; Heritability; Epigenetics

id: bio-behavior-genetics · Unit 1 — Biological Bases of Behavior · standard genetics twins heritability epigenetics methods

Behavior genetics research methods separate genetic influence from environmental influence:

- **Twin studies:** compare **monozygotic (identical)** twins — from one fertilized egg (zygote), genetically identical — with **fraternal (dizygotic)** twins — from two eggs, genetically similar as any siblings. If identical twins raised apart still show highly correlated traits, genetics play a large role. Example: Bouchard’s Minnesota twin studies found IQ correlates strongly even when identical twins are raised apart, though environment has some effect; similar looks may evoke similar treatment, creating a “same effective psychological environment.”
- **Family studies:** examine trait similarity among blood relatives (cannot fully separate shared genes from shared home).
- **Adoption studies:** separate genetic influence (biological parents) from environmental influence (adoptive parents) — the child’s traits compared to both sets of parents.

Heritability: how much of the **variation** in a trait *within a population* is due to genetic differences. It is measured in populations, never individuals, and it can differ between populations and eras.

Misconception: “my IQ is 80% heritable” is a category error — heritability describes variation across a group, not how much of one person’s trait is genetic.

Epigenetics: the study of how the environment (nutrition, stress, toxins, experiences) can alter **gene expression** — switching genes “on” or “off” — *without changing the underlying DNA sequence*. It explains how nurture shapes how nature is expressed (e.g., identical twins developing different conditions); some epigenetic marks can even be passed to offspring.

Exam tip: to attribute a trait mostly to genetics, point to evidence from identical twins (especially raised apart) or adoption designs — not merely to “it runs in the family.”

Chromosomal Abnormalities (Context)

id: bio-chromosomal-disorders · Unit 1 — Biological Bases of Behavior · advanced genetics disorders context

Context beyond the required CED terms: chromosomal-abnormality detail is outside the current CED’s required knowledge, but it remains useful background for Unit 1’s heredity topics — hence this entry is marked *advanced*.

- **Down syndrome:** an extra chromosome on the 21st pair; associated with a rounded face, shorter fingers and toes, slanted wide-set eyes, and intellectual difficulties.
- **Turner’s syndrome:** a female with a single X chromosome; short stature, webbed neck, atypical sexual development.

- **Klinefelter's syndrome:** males with an extra X chromosome (XXY); different sexual development and introversion.
- **PKU (phenylketonuria):** an inherited metabolic condition detectable at birth; dietary management prevents its intellectual effects — a classic example of genes and environment interacting.

Exam tip: you are not required to name these syndromes, but Down syndrome can appear as an example of a chromosomal cause of intellectual disability (connect to the intelligence entries in Unit 2).

The Nervous System

id: bio-nervous-system · Unit 1 — Biological Bases of Behavior

core

nervous-system

cns

pns

neuron

reflex

Central Nervous System (CNS): the brain and spinal cord. The brain receives and processes sensory information, initiates responses, stores memories, and generates thoughts and emotions; the spinal cord conducts signals to and from the brain and produces reflexes. The CNS is the body's decision-making hub.

Peripheral Nervous System (PNS): all nerves outside the brain and spinal cord; relays messages between the CNS and the rest of the body.

- **Somatic nervous system:** controls VOLUNTARY muscle movement — the motor cortex sends impulses through it to the muscles.
- **Autonomic nervous system (ANS):** controls INVOLUNTARY bodily functions — heartbeat, breathing, digestion.
 - **Sympathetic division:** mobilizes the body for stress — “fight or flight.” Dilates pupils, increases heart rate and respiration, stops digestion, releases norepinephrine; involved in excitement and nervousness. Dysfunction is linked to anxiety.
 - **Parasympathetic division:** calms the body after arousal — “rest and digest”; involved in sleep and tiredness. Dysfunction can contribute to insomnia, sleep apnea, and narcolepsy.

Neuron types (by function):

- **Sensory (afferent) neurons:** carry *incoming* information from sensory receptors to the brain and spinal cord.
- **Motor (efferent) neurons:** carry *outgoing* information from the brain and spinal cord to muscles and glands.
- **Interneurons:** found in the spinal cord; relay signals between sensory and motor neurons (sensory-motor integration).

Reflex arc: a quick, involuntary response to stimuli processed at the spinal cord level without waiting for the brain — touching a hot stove: sensory neuron, interneuron in the spinal cord, motor neuron, muscle withdraws the hand. It demonstrates how CNS and PNS neurons work together.

Mnemonic: “SAME” — Sensory = Afferent, Motor = Efferent. Sympathetic = *stress*; parasympathetic = *peace*.

Misconception: the reflex is completed in the spinal cord — the brain is informed afterward, which is why you withdraw your hand before you “feel” the pain consciously.

Note: afferent/efferent terminology itself is not a required CED term, but the neurons of the reflex arc are tested.

Neuron Structure & the Synapse

id: bio-neuron-structure · Unit 1 — Biological Bases of Behavior

core

neuron

anatomy

synapse

glial-cells

Neurons are nerve cells that transmit information — the basic building blocks of the nervous system. **Glial cells** support them by providing structure, insulation, communication, and waste transport.

Neuron anatomy (signal path: dendrites, soma, axon, terminal buttons):

- **Dendrites:** root-like branches that *receive* signals from other neurons; they grow to form new synaptic connections (learning grows dendritic branches).
- **Cell body (soma):** contains the nucleus and the life-support organelles.
- **Axon:** wire-like fiber that carries the outgoing signal from the soma toward the terminal buttons.
- **Myelin sheath:** fatty insulation around the axon that speeds neural impulses; produced by Schwann cells.
- **Nodes of Ranvier:** gaps in the myelin where the action potential “jumps,” speeding transmission on myelinated neurons.
- **Terminal buttons (axon terminals):** branched axon ends that store neurotransmitters in vesicles, ready for release.
- **Synapse (synaptic cleft):** the tiny gap between one neuron’s terminal button and the next neuron’s dendrite — the site of chemical communication between neurons.

Neural transmission is electrochemical: *electrical* within the neuron (dendrites, soma, axon, terminals) and *chemical* between neurons (neurotransmitters cross the synapse; electricity does not jump the gap).

Misconception: the myelin sheath does not decide whether a neuron fires — it only speeds conduction. Deterioration of myelin (as in multiple sclerosis) slows or blocks transmission.

Exam tip: know the direction of the signal. “Receives” points to dendrites; “sends” points to the axon and terminal buttons; the synapse is where neurotransmitters — never electricity — travel.

Neural Firing: The Action Potential

id: bio-neural-firing · Unit 1 — Biological Bases of Behavior · **core** neuron action-potential neural-firing
all-or-none

Neural firing is an **electrochemical** process: electrical within the neuron, chemical between neurons.

Sequence of neural transmission:

1. **Resting potential (polarization):** the inactive neuron holds a slightly negative charge (about -70 mV); negative ions inside, positive ions outside.
2. **Threshold:** the minimum stimulation level needed to trigger firing. If enough excitatory neurotransmitters bind, positive ions rush in.
3. **Action potential (depolarization):** the charge reverses as positive sodium ions rush in; the electrical signal travels down the axon.
4. **All-or-none principle:** a neuron fires completely or not at all — like a gun that cannot be “half-fired.”
5. **Refractory period:** a brief post-firing interval during which the neuron cannot fire again as positive ions pump back out.
6. **Reuptake:** the sending neuron reabsorbs excess neurotransmitter from the synapse.

Intensity coding: a harder slap does not fire neurons *harder* — it fires *more* neurons, and the same neurons fire *more frequently*. Strength of a stimulus is encoded by the number and rate of firing neurons, never by a “bigger” action potential.

Disorders of transmission:

- **Multiple sclerosis (MS):** deterioration of the myelin sheath slows or blocks neural transmission.
- **Myasthenia gravis:** disrupted neurotransmitter (acetylcholine) reception at muscle junctions produces muscle weakness.

Misconception: the action potential is electrical but does not “jump” the synapse — across the synapse the message is chemical.

Note: the sodium-potassium pump is outside the CED scope; know the sequence and the charges conceptually, not the pump mechanics.

Exam tip: “reuptake” belongs to the *sending* neuron; “refractory period” explains why neurons need a brief reset before firing again — both are favorite MCQ distractors.

Neurotransmitters & Hormones

id: bio-neurotransmitters-hormones · Unit 1 — Biological Bases of Behavior · **core** neurotransmitters hormones
endocrine synapse

Neurotransmitters are the chemical messengers that cross the synapse. **Excitatory** transmitters make firing MORE likely; **inhibitory** transmitters make firing LESS likely.

- **Acetylcholine (ACh):** learning, memory, and muscle contraction (including the heart). Too little: Alzheimer’s disease — memory loss and muscle weakness.
- **Dopamine:** movement, learning, attention, emotion, and reward. Too little: tremors, Parkinson’s disease, ADHD. Too much: drug-induced psychosis and schizophrenia (hallucinations and delusions).
- **Serotonin:** mood, hunger, sleep, and arousal. Too little: depression (many antidepressants raise serotonin availability).
- **Norepinephrine:** alertness and arousal (also a stress hormone). Too little: depression; too much: anxiety and high blood pressure.
- **GABA:** the major *inhibitory* transmitter; calms the nervous system. Too little: seizures, anxiety, insomnia, tremors.
- **Glutamate:** the major *excitatory* transmitter; involved in memory. Too much: seizures, migraines, overstimulation, stroke damage.
- **Endorphins:** natural painkillers associated with pleasure; released by stress and exercise. Too little: pain and sadness; artificially elevated (opiates): euphoria and addiction risk.
- **Substance P:** transmits pain signals from the PNS to the CNS.

Mnemonic: GABA is the *brakes*, glutamate is the *gas* — inhibitory vs. excitatory.

Hormones are chemical messengers carried by the bloodstream; they act like neurotransmitters but systemically (body-wide, slower, longer-lasting). The tested hormones: **adrenaline (epinephrine)** — fight-or-flight, from the adrenal glands; **leptin** — satiety (fat cells signal “full”); **ghrelin** — hunger (empty stomach signals “hungry”); **melatonin** — sleepiness, from the pineal gland; **oxytocin** — social bonding, love, and trust.

Pituitary gland: the “master gland,” controlled by the hypothalamus; releases hormones that govern other glands (including growth hormone). The rest of the endocrine glands are beyond the required CED scope.

Misconception: neurotransmitters are not “happy” or “sad” chemicals one-for-one — dopamine is involved in reward AND movement AND attention; effects depend on location, dose, and receptor type.

Exam tip: pair the imbalance with the behavior in the scenario: underactive GABA with a panic presentation; excess dopamine with hallucinations; ACh loss with progressive memory decline.

Psychoactive Drugs & Addiction

id: bio-psychoactive-drugs · Unit 1 — Biological Bases of Behavior · **standard** drugs agonist antagonist
tolerance addiction

Psychoactive drugs work at synapses by stimulating, inhibiting, or mimicking neurotransmitters.

Drug mechanisms:

- **Agonists:** occupy receptors AND activate them — mimic or encourage neural firing (morphine mimics endorphins; nicotine mimics ACh).

- **Antagonists:** occupy receptors but do NOT activate them — block natural neurotransmitters and discourage firing.
- **Reuptake inhibitors:** block reabsorption of neurotransmitters back into the sending cell, prolonging their effect (cocaine; SSRI antidepressants).

Drug categories:

- **Depressants** — reduce neural activity and slow body functions. **Alcohol:** a disinhibitor that slows judgment and inhibitions, disrupts memory by suppressing REM sleep, and reduces self-awareness (expectancy effects also shape behavior; chronic alcohol use disorder shrinks the brain). **Barbiturates/tranquilizers:** sleep and anxiety drugs; lethal when combined with alcohol. **Opiates** (heroin, morphine, codeine, methadone): pain relief; with chronic use the brain stops producing its own endorphins.
- **Stimulants** — excite neural activity and speed body functions (pupils dilate, heart rate rises, appetite drops). **Caffeine; nicotine** (highly addictive; withdrawal brings craving, insomnia, anxiety, irritability); **cocaine** (blocks reuptake of dopamine, serotonin, and norepinephrine — euphoric rush, then agitated depression; cardiovascular stress and suspicion); **amphetamines/methamphetamine** (trigger dopamine release; reduce baseline dopamine over time); **ecstasy/MDMA** (stimulant plus mild hallucinogen; floods serotonin; risks include serotonin neuron damage, immune suppression, dehydration, overheating).
- **Hallucinogens** — distort perception and evoke sensory images without input. **LSD** (“acid”): hallucinations with emotions ranging from euphoria to panic; mood and expectations shape effects. **Marijuana/THC:** mild hallucinogen; can alleviate pain and nausea; risks include impaired attention/learning/memory, traffic accidents, bronchitis, and possible psychosis/anxiety links.
- **Opioids:** pain relievers that mimic endorphins (heroin).

Related processes: **tolerance** — diminishing effect with regular use of the same dose, so the user needs larger doses; **withdrawal** — discomfort and distress after discontinuing an addictive drug; **addiction (substance use disorder)** — continued craving and use despite significant life disruption, graded from mild (2-3 indicators) to severe (6+ indicators of diminished control, social impairment, hazardous use, tolerance/withdrawal).

Misconception: antagonists do not “simulate” a neurotransmitter — they occupy the receptor WITHOUT activating it. An agonist mimics; an antagonist blocks.

Exam tip: identify the mechanism first (agonist / antagonist / reuptake inhibitor), then connect it to the neurotransmitter’s function (e.g., an SSRI as a serotonin reuptake inhibitor leaves more serotonin in the synapse).

Brain Structures & Functions

id: bio-brain-structures · Unit 1 — Biological Bases of Behavior · **core** brain brainstem limbic-system

cerebellum hippocampus

The exam emphasizes brain FUNCTIONS connected to behavior, not structure identification.

Brainstem (hindbrain) — basic survival:

- **Brainstem:** automatic survival functions; the crossover point where nerves cross sides of the body.
- **Medulla:** heartbeat, breathing, and blood pressure.
- **Pons:** coordinates movement and helps regulate sleep (connects hindbrain with midbrain/forebrain). Note: the pons by name is excluded in the newer CED listing, but the content is taught.
- **Reticular activating system (reticular formation):** network in the brainstem controlling arousal, alertness, attention, and the sleep-wake cycle.
- **Reward center:** dopamine-rich circuitry that fires in response to rewarding stimuli; drives motivation.
- **Cerebellum (“little brain,” bottom-rear):** coordination of voluntary muscle movement, balance, and procedural learning of habitual movements (e.g., tracking a target with the eyes).

Limbic system (forebrain) — emotion and drives:

- **Thalamus:** sensory *relay station* on top of the brainstem — directs incoming sensory signals to the correct cortical areas, like air-traffic control. Smell bypasses the thalamus.
- **Hypothalamus:** regulates hunger, thirst, body temperature, sexual arousal, and biological rhythms (wake/sleep); controls the endocrine system via the pituitary; contains the SCN (circadian clock).
- **Pituitary gland:** the “master gland,” controlled by the hypothalamus.
- **Amygdala:** emotion — especially *fear and aggression*.
- **Hippocampus:** memory formation — processes new memories and sends them to the cortex for storage; damage leaves a person unable to form new memories (anterograde amnesia; the famous case H.M.).

Corpus callosum: the large C-shaped band of nerve fibers connecting the two hemispheres — the largest bundle of white matter; allows interhemispheric communication.

Misconception: the amygdala is not only fear — but “fear and aggression” is the pairing the exam expects. And the hippocampus stores nothing itself; it *forms* new memories and hands them to the cortex.

Exam tip: a scenario about a person who cannot form new memories after hippocampal damage, or who is fearless after amygdala damage, should be explained through the structure’s FUNCTION.

The Cerebral Cortex, Language & Split-Brain

id: bio-cortex · Unit 1 — Biological Bases of Behavior · **core** cortex lobes language plasticity split-brain

The **cerebral cortex** is the wrinkled outer “gray matter” layer, densely packed with neurons; the wrinkles (fissures/sulci) increase surface area. It divides into two **hemispheres** with **contralateral organization** — the left hemisphere controls and senses the right side of the body, and vice versa. **Hemispheric specialization:** the left hemisphere is typically more active in logic, language, and sequential tasks; the right in spatial, creative, and intuitive tasks (still under investigation — beware pop “left-brained/right-brained” claims).

The four lobes:

- **Frontal** (behind the forehead): speaking, muscle movements (motor cortex at its rear), planning, judgments, higher-order thinking, and **executive functioning**; contains the **prefrontal cortex** — abstract thought, emotional control, personality.
- **Parietal** (top/rear): the **somatosensory cortex** at its front strip — registers touch, pressure, temperature, and pain, organized top-down (top of the strip = feet, bottom = face/head); also association areas and spatial orientation.
- **Occipital** (very back): visual processing — the visual cortex receives impulses from the retinas.
- **Temporal** (above the ears): auditory processing and language comprehension.

Motor cortex: rear strip of the frontal lobe; plans, controls, and executes voluntary skeletal movement, organized top-down like a body map.

Language areas (typically LEFT hemisphere):

- **Broca’s area** (frontal lobe): speech *production* — controls the muscles for speaking. Damage produces **Broca’s aphasia**: the person understands speech but cannot speak fluently.
- **Wernicke’s area** (temporal lobe): language *comprehension* — interprets written and spoken language. Damage produces **Wernicke’s aphasia**: fluent but meaningless, nonsensical speech with poor understanding.
- **Aphasia:** general impairment of language from left-hemisphere damage. Broca-impaired patients may still *sing* familiar songs.

Split-brain research: patients whose corpus callosum is severed (a treatment for severe epilepsy). Information shown in the *right visual field* goes to the left hemisphere (language) and can be named;

information in the *left visual field* goes to the right hemisphere and cannot be orally reported — though the left hand can point to it. Demonstrates hemisphere specialization and contralateral organization.

Brain plasticity: the brain's ability to rewire, modify, or create new connections throughout development; a damaged function may be assumed by other regions, especially in youth.

Misconception: vision is occipital but hearing is temporal (not occipital); and split-brain patients cannot *say* what they saw in the left visual field — but they are not “unaware” of it.

Studying the Brain

id: bio-brain-research-methods · Unit 1 — Biological Bases of Behavior · [standard](#) [brain](#) [imaging](#) [eeg](#) [methods](#)

How psychologists study the brain:

1. **Accidents and case studies:** in-depth examination of rare brain damage. **Phineas Gage** (1848) — an iron rod through the frontal lobe left him impulsive and emotionally volatile, evidence that the frontal lobe regulates emotion and personality.
2. **Lesioning:** intentional removal or destruction of brain tissue to study function (historically used to calm psychiatric patients).
3. **EEG (electroencephalogram):** electrodes on the scalp record electrical activity and brain waves — measures *function*; widely used in sleep research to identify sleep stages.
4. **CT/CAT scan:** rotating X-ray cameras build a 3-D image of brain *structure*.
5. **MRI:** magnetic fields measure the density and location of brain material — detailed *structure*, no X-rays.
6. **PET scan:** a radioactive glucose tracer shows which areas are most *active* (function) during tasks.
7. **fMRI:** combines MRI and PET — shows *both* structure and function via blood flow.

Misconception: an MRI is not a “function” scan — structure scans (CT, MRI) show anatomy; function scans (EEG, PET, fMRI) show activity.

Exam tip: the exam tests the logic of these tools rather than their names as required terms. If a study “sees which brain areas light up while judging faces,” that is a function scan (PET/fMRI); if it “reveals a tumor’s location,” that is structure (MRI/CT).

Consciousness, Circadian Rhythms & Sleep Stages

id: bio-sleep-circadian-stages · Unit 1 — Biological Bases of Behavior · [core](#) [sleep](#) [consciousness](#) [circadian](#) [rem](#)

Consciousness: varying levels of awareness of thoughts, feelings, behavior, and events in one’s internal and external worlds. **Sleep and wakefulness are two types of consciousness.**

Circadian rhythm: the roughly 24-hour biological clock governing body temperature, arousal, sleeping, and waking; age and experience can reset it. The **suprachiasmatic nucleus (SCN)** — cell clusters in the hypothalamus — controls the circadian rhythm: in response to light, the SCN signals the **pineal gland** to adjust **melatonin** production, modifying sleepiness. **(The SCN/pineal mechanism is taught although flagged beyond the CED’s required terms.)**

Jet lag and shift work are disruptions of the circadian rhythm; night-shift workers risk fatigue, stomach problems, heart disease, and breast cancer (desynchronization). Sleep patterns are also culturally influenced.

Sleep stages (identified by EEG; 90-minute cycles, repeated through the night): deep sleep *diminishes* and REM *increases* across the night.

- **Awake, relaxed:** alpha waves (slow); calm, drowsy state.
- **NREM-1:** irregular theta waves; lightest sleep; **hypnagogic sensations** (jerks, floating/falling feelings) and hallucinations (false sensory experiences); lasts a few minutes; easy to wake.

- **NREM-2: sleep spindles** (bursts of rapid rhythmic activity); about 20 minutes; clearly asleep but easily awakened.
- **NREM-3: delta waves** (large, slow); deep slow-wave sleep (30 minutes); hard to awaken; children spend more time here.
- **REM:** rapid saw-toothed waves (resembling NREM-1); **paradoxical sleep** — the body is fully relaxed (paralyzed) while brain and body systems are active; heart rate rises, breathing irregular; rapid eye movements; *vivid dreams occur here*; **REM rebound** — extra REM after deprivation.

NREM means non-rapid-eye-movement sleep (stages 1-3); its total duration decreases across the night. **Aging and sleep:** older adults awaken more easily and get less NREM-3 deep sleep.

Misconception: REM is not the deepest sleep — it is near-wakefulness in the brain with a paralyzed body. The deepest sleep is NREM-3.

Exam tip: match the stage to the marker: hypnagogic jerks to NREM-1; sleep spindles to NREM-2; delta and bedwetting/night terrors/sleepwalking to NREM-3; vivid dreams and paralysis to REM.

Why We Sleep & Theories of Dreams

id: bio-sleep-functions-dreams · Unit 1 — Biological Bases of Behavior · [standard](#) [sleep](#) [dreams](#)
[memory-consolidation](#)

Functions of sleep:

1. **Protection:** kept our ancestors safe during dangerous night periods.
2. **Recuperation/restoration:** restores and repairs damaged neurons and depleted resources.
3. **Memory consolidation:** replays and strengthens the day's neural connections and memories.
4. **Creative thinking:** promotes next-day creative problem-solving.
5. **Growth:** the pituitary gland releases growth hormone during slow-wave sleep.

Theories of dreams:

1. **Activation-synthesis theory:** REM neural activity evokes random visual memories; the brain *synthesizes* them into a story — a bottom-up interpretation of “neural static.”
2. **Information-processing/consolidation theory:** dreams help sort the day's events and consolidate them into memory (brain scans link REM and memory) — a top-down view.
3. **Physiological function:** REM stimulation develops and preserves neural pathways (infants spend abundant time in REM).
4. **Restoration theory:** sleep restores energy and resources for health and productivity.
5. **Cognitive development:** dreams reflect the dreamer's knowledge and cognitive development, simulating reality — including worst-case scenarios.

(Freud's wish-fulfillment / manifest-latent dream view is **NOT** on the exam.)

Sleep deprivation effects: fatigue, irritability, impaired concentration and productivity, impaired memory consolidation, immune suppression, weight gain, depression, joint pain, and accident vulnerability.

Misconception: dreams are not proved to have symbolic hidden meanings — the tested theories are physiological and cognitive, not psychoanalytic.

Exam tip: “memory consolidation” doubles as a sleep FUNCTION and a dream THEORY — read the question stem to see which is being asked.

Sleep Disorders & Deprivation

id: bio-sleep-disorders · Unit 1 — Biological Bases of Behavior · [standard](#) [sleep](#) [disorders](#) [insomnia](#) [rem](#)

Sleep disorders (the tested set):

- **Insomnia:** recurring problems falling or staying asleep — the most common sleep disorder.

- **Narcolepsy:** uncontrollable daytime sleep attacks; the person may lapse directly into REM sleep.
- **Sleep apnea:** temporary breathing cessations during sleep with repeated micro-awakenings; associated with obesity (especially in men); causes daytime fatigue.
- **REM sleep behavior disorder:** acting out vivid dreams physically or vocally (“dream-enacting”) because the normal REM paralysis fails.
- **Somnambulism (sleepwalking):** occurs in *NREM-3*; the person shows normal waking activities while asleep; children rarely remember it; few serious concerns.
- **Night terrors:** high arousal and an appearance of terror during early-night *NREM-3*; heart and breathing rates double; children remember little; rare in adulthood.

Misconception: night *terrors* are NREM-3, early night; night *mares* are bad dreams in morning REM. Sleepwalking is NOT “acting out dreams” — that is REM sleep behavior disorder; sleepwalking happens in deep NREM-3 when vivid dreaming is unlikely.

Exam tip: the stage linkage is the testable hook — NREM-3: sleepwalking, night terrors, delta waves; REM: vivid dreams, paralysis, REM rebound, narcolepsy’s direct entry.

Sensation: Basic Principles

id: bio-sensation-basics · Unit 1 — Biological Bases of Behavior · **core** sensation transduction thresholds
webers-law

Sensation: the process of detecting environmental stimuli that meet thresholds and *transducing* them into neural messages. (**Perception** — organizing and interpreting that information — belongs to Unit 2.1.)

Three steps of sensation: (1) receive stimulation via receptor cells; (2) transform (transduce) it into neural impulses; (3) deliver the impulses to the brain.

- **Transduction:** conversion of one form of energy (light, sound, chemicals) into neural impulses the brain can interpret.
- **Absolute threshold:** the minimum stimulation detected *50% of the time*.
- **Difference threshold (just noticeable difference, JND):** the minimum *difference* between two stimuli detectable 50% of the time.
- **Weber’s law:** two stimuli must differ by a constant minimum *percentage* (not amount) to be perceived as different — 1 extra ounce is noticeable on a 10-ounce weight, not on a 100-ounce weight.
- **Signal detection theory:** detection depends on experience, expectations, motivation, and alertness — there is no single absolute threshold. (**Taught though flagged outside the CED’s required terms.**)
- **Sensory adaptation:** diminished sensitivity to unchanging stimuli as nerve cells fire less — no longer noticing perfume after a few minutes.
- **Sensory interaction:** one sense influences another — smell plus texture plus taste equals flavor; the McGurk effect; food tastes bland with a blocked nose.
- **Synesthesia:** stimulation of one sense triggers an experience in another — “seeing” sounds as colors.

Misconception: sensory adaptation is not “getting used to” pain in the everyday sense — most pain stimuli do NOT adapt; unchanging, harmless stimuli (pressure, odor) do.

Exam tip: absolute vs. difference thresholds both cite 50% detection — the difference is whether the question is about detecting a STIMULUS or detecting a CHANGE between stimuli.

Vision

id: bio-vision · Unit 1 — Biological Bases of Behavior · **core** vision color retina perception

Eye pathway: cornea (protective outer layer) — pupil (adjustable opening) controlled by the iris (colored muscle) — lens (focuses light via **accommodation**, changing shape for near/far objects) — retina (light-sensitive inner surface) — photoreceptors — bipolar cells — ganglion cells — optic nerve — thalamus — occipital cortex. **Accommodation failure: nearsightedness** (light focuses in front of the retina) and **farsightedness** (behind the retina).

Photoreceptors:

- **Rods:** detect black, white, and gray; detect movement; handle peripheral and twilight/low-light vision; light/dark adaptation.
- **Cones:** cluster in the **fovea** (the point of central focus); detect fine detail and *color*; require good light. Blue = short wavelengths, green = medium, red = long.
- **Fovea:** contains only cones; maximum visual acuity — sharpest vision for reading and driving.
- **Blind spot:** the point where the optic nerve exits the eye; no receptor cells there; the brain fills the gap (relates to the Gestalt principle of closure).
- **Feature detector cells:** visual cortex neurons that fire to specific features — edges, lines, angles, movement. (Named term flagged beyond the CED's required list.)
- **Parallel processing:** the brain simultaneously processes subdimensions of a visual scene — motion, form, depth, color.

Color vision theories (both are correct, at different stages):

1. **Trichromatic (Young-Helmholtz) theory:** three cone types (red/long, green/medium, blue/short wavelengths) combine to produce all colors; explains processing at the *retina*.
2. **Opponent-process theory:** opposing retinal/ganglion-cell processes (red-green, blue-yellow, black-white) enable color vision; explains *afterimages* (stare at green, see red).

Color vision deficiency: damage or irregularity in cones or ganglion cells. **Dichromatism:** one cone type nonfunctioning — cannot distinguish one color pair (e.g., red-green). **Monochromatism:** only one (or no) cone type functioning — the world in shades of gray.

Visual disorders (brain-based, occipital-lobe damage): **prosopagnosia** — face blindness: can see faces but cannot recognize identities; **blindsight** — reported “blindness” with some intact visual responses.

Misconception: the retina's rods handle peripheral vision, which is why dim stars vanish when you look straight at them (cones need light); and color processing starts trichromatic in the cones but is opponent-process by the ganglion cells.

Exam tip: afterimage questions = opponent-process; “three cone types” questions = trichromatic. Know which theory explains which evidence.

Hearing (Audition)

id: bio-hearing · Unit 1 — Biological Bases of Behavior · [standard](#) [hearing](#) [audition](#) [cochlea](#) [pitch](#)

Sound waves are compressed and expanded air. **Wavelength/frequency determines pitch** (the experienced highness/lowness — long waves = low pitch); **amplitude determines loudness** (measured in decibels).

Ear pathway: outer ear (pinna + auditory canal) — **eardrum (tympanic membrane)** vibrates — **middle-ear bones** (hammer/malleus, anvil/incus, stirrup/stapes) amplify — **oval window** — **cochlea** (coiled, fluid-filled) ripples the **basilar membrane**, bending **hair cells** — impulses travel the **auditory nerve** — thalamus — auditory cortex (temporal lobe). (**Ear anatomy beyond the CED-listed structures is context.**)

Pitch theories:

1. **Place theory:** different pitches vibrate different *places* along the basilar membrane; explains *high* pitches only (low pitches vibrate too broadly).

2. **Frequency (temporal) theory:** the rate of nerve impulses matches the sound's frequency; the whole membrane vibrates; limited because neurons cannot fire more than about 1000 times per second.
3. **Volley principle:** neural cells alternate firing to collectively match frequencies above an individual neuron's limit — extends frequency theory into mid/high ranges.

Together: place theory handles high pitches; frequency plus volley handles low pitches; a combination explains intermediate pitches.

Sound localization: sound strikes one ear sooner or more intensely; the brain computes the source from the inter-aural difference (creating a “sound shadow”).

Hearing loss:

- **Conduction deafness:** damage to the *mechanical* system (eardrum, middle-ear bones); less common.
- **Sensorineural (nerve) deafness:** damage to cochlea hair cells or the auditory nerve; the most common form; the person may hear sound but cannot discern speech; treated with **cochlear implants** — bionic devices that translate sound into electrical signals wired into the cochlea's nerves.

Misconception: a cochlear implant is not a hearing aid that amplifies sound — it bypasses damaged hair cells and directly stimulates the auditory nerve.

Exam tip: “grandmother can hear you but cannot understand your words” = sensorineural deafness (hair-cell damage), the classic scenario.

Smell & Taste (Chemical Senses)

id: bio-chemical-senses · Unit 1 — Biological Bases of Behavior · [standard](#) [smell](#) [taste](#) [olfaction](#)
[chemical-senses](#)

Olfaction (smell): odor molecules reach roughly 20 million receptor cells atop each nasal cavity, then the **olfactory bulb**, then the brain. **Smell is the only sense NOT processed first through the thalamus** — it routes to the amygdala and hippocampus, which is why smells trigger emotion and memory so powerfully. **Anosmia** is the inability to smell. **Pheromones** are chemical signals (sexual attractants) detected by other members of a species. Women and young adults have the keenest smell.

Gustation (taste): **taste buds** sit on **papillae** and house receptor cells; the sensations are **sweet, sour, salty, bitter, umami, and oleogustus**. Taste receptors regenerate every 1-2 weeks; they decline with age, smoking, and alcohol. **Supertasters/medium tasters/non-tasters** are classified by receptor density (linked to taste sensitivity). Smell and taste interact strongly — without smell, taste is muted (sensory interaction).

Misconception: “taste” is mostly flavor, and flavor is multisensory — the tongue detects only the basic qualities above; much of what we call taste is retronasal smell plus texture.

Exam tip: the thalamus exception is a favorite question: every sense relays through the thalamus EXCEPT smell, which connects directly to emotion/memory circuitry.

Touch & Pain

id: bio-touch-pain · Unit 1 — Biological Bases of Behavior · [standard](#) [touch](#) [pain](#) [mechanoreceptors](#)
[gate-control](#)

Touch: the skin's four basic sensations are **pressure, warmth, cold, and pain**; mixtures create other experiences — cold plus pressure equals *wetness*; adjacent warm and cold spots equal *hot*.

Skin receptor types (each transduces a different form of energy):

- **Mechanoreceptors:** transduce physical force — pressure, vibration, texture, and stretch — into neural signals; the basis of pressure/touch perception.
- **Thermoreceptors:** transduce temperature; separate receptors for warmth and cold.
- **Nociceptors:** detect hurtful temperatures, pressures, and chemicals (in skin, muscles, and organs) — the pain receptors.

Cognition shapes sensation: tickling yourself activates the somatosensory cortex less than another person's tickle — the brain predicts and dampens self-produced touch.

Pain:

- **Gate-control theory:** the spinal cord contains a neurological “gate.” Pain signals traveling on *small* nerve fibers OPEN the gate (pain felt); *large*-fiber activity (massage, electric stimulation, acupuncture) or brain messages CLOSE the gate (pain blocked). This is why “gate-closing” stimulation and distraction are used to treat pain.
- **Phantom limb sensation:** pain or sensation in amputated limbs — the brain misinterprets and amplifies spontaneous CNS activity; we feel with our *brain*, not our limb.
- **Endorphins:** natural painkillers; placebos and distraction also modulate pain.
- **Pain memory** records the *peak* and the *end* of pain, not the average.

Misconception: nociceptors are not “pain located in the skin only” — they also monitor muscles and organs; and pain intensity is shaped top-down by attention, expectation, and mood, not just by tissue damage.

Exam tip: gate-control questions hinge on fiber size: small fibers open the gate (more pain), large fibers close it (less pain) — match the intervention (massage, distraction, acupuncture) to gate-CLOSING.

Body Position & Movement

id: bio-vestibular-kinesthetic · Unit 1 — Biological Bases of Behavior · [standard](#) [vestibular](#) [kinesthesia](#) [balance](#)

Vestibular sense (balance): monitors head position and movement via fluid in the **semicircular canals** and **vestibular sacs** of the inner ear; signals go to the cerebellum, producing super-speedy reflexive balance corrections. Damage produces dizziness and loss of balance (e.g., spinning until the fluid lags and the world seems to move).

Kinesthesia: the sense of the position and movement of *individual body parts*, via millions of sensors in muscles, tendons, and joints. Vision interacts with it — touch your nose with your eyes closed and kinesthesia guides your finger.

Distinction: vestibular = the WHOLE body's balance and head motion (inner ear); kinesthetic = individual PARTS in relation to each other (muscles, tendons, joints).

Misconception: kinesthesia is not “proprioception of pain” or touch — it is position sense, and it works without vision; the vestibular sense is not in the joints but in the inner ear.

Exam tip: scenario cues — “dizzy after the fair ride” = vestibular; “athlete adjusts limb position mid-air without looking” = kinesthetic.

Unit 2 — Cognition

Perception: Processing, Schemas & Attention

id: cog-perception-principles · Unit 2 — Cognition · [core](#) [perception](#) [gestalt](#) [attention](#) [top-down](#)

Bottom-up processing: analysis starting at the sensory receptors and building up to the brain's integration — raw, data-driven. **Top-down processing:** interpretation guided by prior knowledge, experience, and expectations — concept-driven (“needs thinking!”).

Schemas and expectations:

- **Schema:** a concept or framework that organizes and interprets information. (Schemas change through assimilation and accommodation — see the Piaget entry.)
- **Perceptual set:** a mental predisposition to perceive one thing and not another — top-down expectations filter perception (hearing “the rat ate the cat” in backwards music only after being told to look for it).
- **Context effects:** the environment or situation influences thinking, feeling, and perception — a “suit” means a swimsuit at the beach.

Gestalt principles: the whole differs from the sum of its parts; the brain organizes stimuli into meaningful wholes.

- **Figure-ground:** organizing the visual field into objects (figures) that stand out from their surroundings (ground).
- **Closure:** filling in gaps to perceive complete objects (also how the brain fills the blind spot).
- **Proximity:** nearby elements are grouped together.
- **Similarity:** similar elements are perceived as a group.
- **Continuity:** smooth, continuous patterns are preferred over discontinuous ones.

Attention:

- **Selective attention:** focusing conscious awareness on one particular stimulus; attention is limited (hence the danger of texting while driving).
- **Cocktail party effect:** attending to one voice in a noisy room — especially hearing your own name.
- **Inattentional blindness:** failing to see visible objects when attention is elsewhere (the invisible gorilla).
- **Change blindness:** failing to notice environmental changes; a form of inattentional blindness.

Misconception: inattentional blindness is not a vision defect — the image lands on the retina; attention, not the eye, fails.

Exam tip: scenario cues — expecting to see the doctor = perceptual set; grouping co-located dots = Gestalt proximity; missing the moonwalking bear = inattentional blindness.

Perception: Depth, Constancy & Motion

id: cog-perception-depth · Unit 2 — Cognition · **core** perception depth binocular monocular illusions

Depth perception: seeing in three dimensions from two-dimensional retinal images. The **visual cliff** (Gibson and Walk): infants crawl to the “shallow” side but avoid the glass-covered “cliff” side — depth perception appears innately or very early.

Binocular cues (require two eyes):

- **Retinal disparity:** the slightly different images from each eye (about 2.5 inches apart); the brain compares them — greater disparity = closer object.
- **Convergence:** the eyes turn inward for near objects; the more the eyes converge, the closer the object (muscle feedback).

Monocular cues (one eye alone; only these are tested):

- **Relative clarity:** clearer objects seem closer; hazy ones seem farther.
- **Relative size:** if two objects are assumed similar in size, the smaller retinal image seems farther.
- **Texture gradient:** texture grows denser and finer-grained with distance.
- **Linear perspective:** parallel lines converge toward a vanishing point at the horizon.
- **Interposition:** one object partially blocking another is perceived as closer.

(Relative height and motion parallax are taught but flagged outside the CED.)

Perceptual constancy: perceiving objects as unchanging (consistent size, shape, color, lightness) despite changing retinal images — a door stays a rectangle even viewed at an angle; **color/brightness constancy** depends on surrounding context; **size constancy** keeps familiar objects constant across distances.

Motion and illusions: phi phenomenon / apparent motion — adjacent lights blinking in quick succession look like movement (marquee signs); **stroboscopic movement** — rapid series of slightly varying images looks continuous (film). **Moon illusion:** the moon looks larger near the horizon because monocular distance cues make it seem farther, so the brain perceives it as larger. **Perceptual adaptation:** the ability to adjust to changed or inverted visual input (distortion goggles) within minutes. **Culture and perception:** susceptibility to the Mueller-Lyer illusion varies with exposure to “carpentered” (rectangular) environments.

Misconception: retinal disparity is about comparing TWO images; convergence is about MUSCLE feedback — both binocular; all five tested monocular cues work with one eye covered.

Exam tip: name the cue from the drawing — overlapped objects = interposition; converging railroad tracks = linear perspective; hazy mountains = relative clarity.

Thinking, Problem-Solving & Decision-Making

id: cog-thinking-decisions · Unit 2 — Cognition · core thinking heuristics problem-solving creativity

Concepts: mental groupings of similar objects, events, ideas, or people — the building blocks of thought. **Prototypes:** the best or ideal mental example of a category (a robin for “bird”). **Schemas** are frameworks modified through **assimilation** (interpreting new experiences within existing schemas) and **accommodation** (adjusting schemas to fit new information).

Problem-solving strategies:

- **Algorithm:** a methodical, logical rule or procedure that GUARANTEES a solution — slow but certain (trying every combination).
- **Heuristic:** a simple thinking shortcut allowing quick judgments but prone to error.
 - **Representativeness heuristic:** judging likelihood by similarity to a prototype (assuming a quiet, bookish person is a librarian, not a farmer).
 - **Availability heuristic:** estimating likelihood by how easily examples come to mind (fearing plane crashes over car crashes because crashes are vivid and reported).

Obstacles to problem-solving:

- **Mental set:** approaching problems with previously successful methods even when better options exist.
- **Functional fixedness:** perceiving objects only in their usual functions — a classic creativity block (the participant who generated zero uses for tweezers).
- **Framing:** the way an issue is posed changes decisions (“90% survival” vs. “10% mortality”); ties to the misinformation effect.
- **Priming:** unconscious activation of associations that predisposes perception, memory, or response (“yellow” speeds recognition of “banana”).
- **Gambler’s fallacy:** believing random events are influenced by previous outcomes (“I’m due for heads”).
- **Sunk-cost fallacy:** continuing an endeavor because of resources already invested.

Thinking about thinking: metacognition — awareness and monitoring of one’s own cognitive processes, including memory strategies. **Executive functions:** cognitive processes that let individuals generate, organize, plan, and carry out goal-directed behaviors and critical thinking.

Creativity: producing novel, valuable ideas. **Divergent thinking** expands possibilities and generates many solutions; **convergent thinking** narrows to the single best solution; **insight** is the sudden realization of a solution — the “aha” moment.

Misconception: heuristics are not “lazy thinking” gone wrong — they are usually fast and right; the named errors show their predictable failure modes.

Exam tip: availability = “how easily can I recall examples?”; representativeness = “how well does this match my prototype?” Match the scenario’s cue to the right shortcut.

Memory: The Information-Processing Model

id: cog-encoding-storage-retrieval · Unit 2 — Cognition · core memory encoding storage retrieval

working-memory

Memory: the persistence of learning over time through three processes — **encoding** (getting information in), **storage** (retaining it), and **retrieval** (getting it out).

Information-processing model: memory works like a computer — encode, store, retrieve. **Automatic processing** encodes incidental information unconsciously (space, time, frequency); **effortful processing** requires attention and conscious work.

Memory types:

- **Explicit (declarative) memory:** conscious memories of facts and experiences, easily described — **episodic memory** (personal experiences and events) and **semantic memory** (facts and general knowledge).
- **Implicit (nondeclarative) memory:** retention without conscious recollection — **procedural memory** for skills and operations (riding a bike).
- **Prospective memory:** remembering to perform planned actions at future moments (taking medication at 8 p.m.).

Neural basis: long-term potentiation (LTP) — the persistent strengthening of synapses based on recent activity patterns; receiving neurons become more efficient (more receptors). LTP is the neural basis of learning and memory.

Multi-store (Atkinson-Shiffrin) model: sensory memory, then short-term memory, then long-term memory.

- **Sensory memory:** immediate, brief recording of sensory information — **iconic memory** (visual, a few tenths of a second) and **echoic memory** (auditory, 3-4 seconds).
- **Short-term memory (STM):** briefly holds about 7 items before storage or loss (20-30 seconds without rehearsal).
- **Long-term memory (LTM):** relatively permanent, limitless storage.

Working memory model (Baddeley): an updated view of STM as ACTIVE processing of incoming and retrieved information, with a limited capacity for simultaneous processing:

- **Central executive:** directs attention and processing.
- **Phonological loop:** verbal/auditory information.
- **Visuospatial sketchpad:** visual/spatial information.
- **Episodic buffer:** integrates the streams (taught).

Levels of processing: retention depends on processing DEPTH — **structural** (physical features, shallowest), **phonemic** (sound), **semantic** (meaning — deepest, best recall).

Misconception: short-term memory and working memory are not synonyms on the exam — STM stores briefly; working memory actively manipulates information with named components.

Exam tip: map the scenario to the model: “recalling a phone number just long enough to dial” = STM; “mentally rotating furniture in a room” = visuospatial sketchpad; “knowing Paris is the capital of France” = explicit/semantic.

Encoding Memories: Mnemonics & Spacing

id: cog-encoding-strategies · Unit 2 — Cognition · core memory encoding mnemonics spacing-effect

Mnemonic devices: memory aids — especially techniques using vivid imagery and organizational cues (acronyms, the peg-word method, the method of loci).

- **Method of loci:** associating items with familiar locations along a mental route — walk the route and “pick up” each item.
- **Chunking:** organizing items into familiar, meaningful units (phone numbers as 3-3-4 digits) — expands working memory capacity.

Spacing effect: distributed practice (spreading study over time) beats **massed practice** (cramming) for long-term retention. Ten minutes a day across a week outperforms seventy minutes the night before.

Serial position effect: best recall for the BEGINNING and END of a list.

- **Primacy effect:** better recall for first items — rehearsed into long-term memory.
- **Recency effect:** better recall for last items — still sitting in working memory.

Misconception: chunking does not increase the capacity limit itself — it repackages items so FEWER chunks carry the same information; and the recency effect is not evidence of long-term storage (it vanishes after a delay, while primacy survives).

Exam tip: for “how should a student study?” questions, the scoring pair is distributed practice over massed practice, ideally with deep semantic processing and retrieval practice — connect the strategy to the mechanism (encoding depth, LTP).

Storing Memories: Systems & Impairments

id: cog-storing-memories · Unit 2 — Cognition · standard memory storage amnesia flashbulb

Four storage systems differ in duration, capacity, and content: sensory memory, short-term memory, working memory, and long-term memory.

Rehearsal types:

- **Maintenance rehearsal:** simple repetition keeps information in STM — shallow processing.
- **Elaborative rehearsal:** linking new information to existing knowledge promotes transfer to LTM — deep processing.

Personal memory: autobiographical memory — memory for one’s own life; self-relevant memories are more memorable. **Flashbulb memory:** an unusually clear, vivid memory of an emotionally significant or momentous event (where you were during a national tragedy) — strong emotional arousal “photographs” the moment; notably, flashbulb memories are NOT necessarily more accurate than ordinary memories. **Highly superior autobiographical memory (HSAM):** the rare ability to recall detailed personal experiences for nearly every day of one’s life — evidence of the biological bases of superior memory.

Memory impairments:

- **Retrograde amnesia:** cannot retrieve memories from BEFORE the injury.
- **Anterograde amnesia:** cannot form NEW memories after the injury (patient H.M.; hippocampal damage).
- **Alzheimer’s disease:** progressive, irreversible deterioration of memory, reasoning, language, and physical functioning (linked to acetylcholine loss).
- **Infantile amnesia:** the inability to retrieve memories from much before age 3.

Misconception: H.M.’s profile is the classic discrimination — he could not form NEW explicit memories (anterograde) yet could learn new skills (implicit/procedural memory intact), because the hippocampus supports explicit encoding, not skill learning.

Exam tip: “before or after” is the whole question for amnesias — retrograde loses the PAST; anterograde loses the FUTURE (new learning).

Retrieving Memories

id: cog-retrieving-memories · Unit 2 — Cognition · **core** memory retrieval testing-effect cues

Retrieval cues: anchors that activate associations, aiding recall; **priming** also aids retrieval by activating associations beneath awareness.

Three measures of retention:

- **Recall:** retrieving information **WITHOUT** cues (fill-in-the-blank, essays) — hardest.
- **Recognition:** identifying previously learned information **WITH** cues (multiple choice) — easier.
- **Relearning:** learning again takes less time than the first time (the savings method) — most sensitive.

Retrieval improves when conditions match encoding:

- **Context-dependent memory:** the same **ENVIRONMENT** prompts better recall (scuba divers recall better underwater).
- **State-dependent memory:** the same **PHYSICAL** or physiological state (drug state, mood) prompts better recall.
- **Mood-congruent memory:** we recall experiences **CONSISTENT WITH OUR CURRENT MOOD** (a depressed mood surfaces negative events).

Retrieval practice: actively retrieving information strengthens memory — **the testing effect:** enhanced memory after **RETRIEVING** information rather than rereading it. Testing **IS** studying.

Misconception: rereading and highlighting feel productive but produce weak, shallow traces; self-quizzing feels harder precisely because it builds retrieval strength.

Exam tip: distinguish the three “matching” effects by their match — environment = context-dependent; body/drug state = state-dependent; emotional tone of the **CONTENT** = mood-congruent.

Forgetting & Other Memory Challenges

id: cog-forgetting · Unit 2 — Cognition · **standard** forgetting memory misinformation interference

Forgetting curve (Ebbinghaus): forgetting is rapid immediately after learning, then levels off.

Why we forget:

- **Encoding failure:** the information never entered long-term memory (the penny-drawing task — most people cannot draw a penny’s details).
- **Storage decay:** memory traces fade over time from disuse.
- **Retrieval failure:** the information **IS** stored but inaccessible — e.g., the **tip-of-the-tongue phenomenon**: knowing the information exists but being temporarily unable to retrieve it.

Interference (both are retrieval failure):

- **Proactive interference:** **OLD** information disrupts recall of **NEW** information (last year’s password blocks the new one).
- **Retroactive interference:** **NEW** information disrupts recall of **OLD** information (the new phone number wipes the old one).

Motivated forgetting: repression — in psychodynamic theory, anxiety-arousing memories are banished from consciousness to protect the ego; a defense mechanism.

Memory construction errors:

- **Misinformation effect:** incorporating misleading information into one’s memory of an event (Loftus’s car-crash studies — “smashed” vs. “hit” changed speed estimates); ties to framing.

- **Source amnesia:** attributing a memory to the wrong source — you experienced the idea but forgot where it came from.
- **Constructive memory:** memories are CONSTRUCTED — rebuilt at each retrieval, not perfectly replayed.
- **Imagination inflation:** repeatedly imagining fake events can create false memories.
- **Deja vu:** the eerie sense that “I’ve experienced this before.”

Mnemonic: proactive = the Past interferes forward (Proactive/Past); retroactive = the Recent interferes backward (Retroactive/Recent).

Exam tip: Loftus-style scenarios (leading questions after an event) = misinformation effect; “mixed up old and new class schedules” = interference — decide which system the scenario targets.

Intelligence: Theories

id: cog-intelligence-theories · Unit 2 — Cognition · **core** intelligence g-factor multiple-intelligences eq

Intelligence: the ability to learn from experience, solve problems, and use knowledge to adapt to new situations.

Theories of intelligence:

- **g — Spearman’s general intelligence:** a single general factor underlying all specific mental abilities (people who score high on one ability tend to score high on others). “Multiple abilities vs. g” remains a live debate.
- **Multiple intelligences (Gardner — tested by concept, not necessarily by name):** intelligence comprises multiple abilities beyond school smarts — linguistic, logical-mathematical, spatial, musical, bodily-kinesthetic, intrapersonal, interpersonal, naturalistic.
- **Triarchic theory (Sternberg — by concept):** three components — **analytical, creative, and practical** intelligence.
- **Emotional intelligence (EQ):** perceiving, understanding, managing, and using emotions — four skills that predict social and workplace success.

Fluid vs. crystallized intelligence:

- **Fluid intelligence:** speedy, abstract reasoning; **DECLINES** in late adulthood (slower neural processing).
- **Crystallized intelligence:** accumulated knowledge and verbal skills; **INCREASES** or stays stable with age.

Research using both cross-sectional and longitudinal studies supports this pattern.

Misconception: “multiple intelligences” does not mean eight separate IQ scores — it challenges the very idea of one general factor; and a decline in fluid intelligence does not mean older adults “know less” (crystallized knowledge keeps growing).

Exam tip: match the scenario to the theory — a brilliant cellist with weak academics = Gardner; a street-smart negotiator = Sternberg’s practical; a calm manager reading the room = EQ; puzzle-speed slowing with age while vocabulary grows = fluid down, crystallized up.

Measuring Intelligence: Tests, Reliability & Validity

id: cog-intelligence-testing · Unit 2 — Cognition · standard intelligence testing validity reliability iq

Measuring intelligence:

- **Historical IQ:** mental age divided by chronological age, times 100.
- **Modern IQ:** standardized with mean = 100 and SD = 15 (deviation IQ); about 68% of test-takers score between 85 and 115.

- **Standardization:** administering tests with consistent procedures and environments; defining meaningful scores by comparison with a pretested (norming) group.
- **Norming:** establishing score distributions from a representative sample.

Reliability — consistency of results:

- **Test-retest:** the same people, the same test, twice — similar results.
- **Split-half:** odd questions vs. even questions agree.

Validity — the test measures what it claims to measure:

- **Construct validity:** the test measures the intended theoretical trait.
- **Content validity:** the test adequately samples the whole domain it claims to measure (an AP exam samples the entire course, not one unit; a driving test includes actual driving).
- **Predictive validity:** the test predicts the criterion behavior (e.g., school success).

Test types: **achievement tests** measure what a person HAS LEARNED; **aptitude tests** predict FUTURE performance or capacity to learn.

Ability ranges: **intellectual disability** — limited mental ability (typically IQ around 70 or below) PLUS difficulty with adaptive, independent everyday functioning (conceptual, social, practical skills), present during the developmental period; causes can be chromosomal or genetic (Down syndrome) or environmental (prenatal alcohol exposure). **Giftedness:** exceptionally high intellectual ability or creativity (commonly IQ of 130 or above — the top 2%), often identified through standardized testing to provide enriched educational services.

Misconception: a reliable test is not necessarily valid — a bathroom scale that always reads 10 pounds heavy is perfectly reliable but invalid; and “gifted” programs identify high ability, not worth.

Exam tip: reliability = CONSISTENCY; validity = ACCURACY. Modern educational uses include identifying students who need services — both remedial and enrichment.

Intelligence: Social & Cultural Issues

id: cog-intelligence-issues · Unit 2 — Cognition · [standard](#) [intelligence](#) [stereotype-threat](#) [flynn-effect](#) [mindset](#)

Social and cultural issues in intelligence testing:

- **Stereotype threat:** anxiety about confirming a negative stereotype about one’s group depresses test performance — the fear itself consumes working memory.
- **Stereotype lift:** the performance boost that comes from positive group stereotypes.
- **Flynn effect:** IQ scores worldwide have risen over generations — attributed to nutrition, health care, education, and smaller families.
- **Within-group vs. between-group variation:** IQ varies MORE WITHIN groups than BETWEEN groups; group-average differences do not explain any individual’s score. Poverty, discrimination, and educational inequity depress scores.
- **Discriminatory uses:** IQ tests have historically limited access to jobs, military ranks, education, and immigration.
- **Fixed vs. growth mindset:** believing intelligence is fixed at birth vs. malleable through effort — mindsets measurably affect academic achievement.

Misconception: the Flynn effect does not mean people are “smarter” in some absolute sense — it means raw scores on IQ tests have risen, forcing periodic re-norming; and stereotype threat is situational, not a trait of any group.

Exam tip: pairing questions are common — stereotype threat (negative stereotype, worse performance) vs. stereotype lift (positive stereotype, better performance); fixed mindset (avoid challenges) vs. growth mindset (embrace effort).

Unit 3 — Development and Learning

Developmental Themes & Research Designs

id: dev-themes-methods · Unit 3 — Development and Learning · **core** development methods longitudinal cross-sectional

Three enduring issues in developmental psychology:

1. **Nature and nurture:** how genetic inheritance and experience interact.
2. **Continuity and stages:** whether development is gradual and continuous or abrupt and stage-like.
3. **Stability and change:** do early traits endure, or do we change as we age?

Research designs:

- **Cross-sectional study:** compares people of DIFFERENT AGES at ONE POINT IN TIME — quick and cheap, but confounded by cohort effects (age differences may really be generational differences; the term is taught although flagged outside the CED).
- **Longitudinal study:** follows and retests the SAME PEOPLE OVER TIME — reveals true within-person change, but slow, expensive, and vulnerable to attrition (participants dropping out).

Misconception: a cross-sectional finding that 70-year-olds score lower than 30-year-olds may reflect education differences between generations (cohort), not aging — that is exactly why longitudinal designs matter for aging research.

Exam tip: identify the design from the timing — “compares groups of different ages now” = cross-sectional; “the same children tested every two years” = longitudinal. Stability vs. change questions want evidence about whether early traits predict later ones.

Physical Development Across the Lifespan

id: dev-physical-development · Unit 3 — Development and Learning · **core** development teratogens puberty motor

Prenatal development: teratogens — chemicals, viruses, and drugs (such as alcohol) that reach the embryo or fetus and cause harm. **Fetal alcohol syndrome** produces facial disproportions and cognitive deficits. Maternal illness, genetic mutations, and hormonal and environmental factors also influence physical and psychological milestones. (**Zygote/embryo/fetus stage names are outside the CED scope.**)

Infancy and childhood:

- **Maturation:** biological growth processes enabling orderly behavioral change, relatively uninfluenced by experience.
- **Motor development:** **fine** motor (small muscles — grasping) and **gross** motor (large muscles — walking) coordination develop in generally the same ORDER with variable TIMING.
- **Infant reflexes:** inborn automatic responses indicating on-track development — **rooting** (cheek touched, head turns and mouth opens toward the stimulus), **sucking**, **grasping** (palm touched, fingers close).
- **Neuroplasticity and pruning:** the brain forms new connections in response to learning; unused connections are eliminated (“pruned”) to strengthen used ones.
- **Critical/sensitive periods:** optimal windows when exposure to certain stimuli produces proper development — especially for language; missing them can permanently limit acquisition (language exposure by about age 7).
- **Imprinting:** in non-human animals, attachment to the first moving object encountered — a means of survival.
- **Visual cliff:** the apparatus demonstrating infants’ early depth perception.

Adolescence: the **adolescent growth spurt** and **puberty** — development of reproductive ability. **Primary sex characteristics** are the reproductive body structures; **secondary sex characteristics** are nonreproductive traits (breasts/hips, voice quality, body hair). **Menarche** = first menstrual period; **spermarche** = first ejaculation. Sex hormones: **estrogens** (females), **testosterone** (males). **Sex** is biological status (chromosomes, anatomy); **gender** is socially constructed expectations.

Adulthood and aging: general leveling-off then decline in reproductive ability (**menopause** — the end of menstruation), mobility, flexibility, reaction time, and visual/auditory acuity; **dementia** and cognitive decline in later life.

Misconception: reflexes like rooting disappear as the cortex matures — their persistence past the expected window signals developmental concern, not health.

Exam tip: teratogen questions stress **TIMING** and **DOSE** — the same exposure harms more during specific prenatal windows; alcohol has no known safe amount in pregnancy.

Gender & Sexual Orientation

id: dev-gender-orientation · Unit 3 — Development and Learning · standard gender sexual-orientation development

Sex is biological status — chromosomes and anatomy. **Gender** is the socially and culturally constructed expectations of boy/girl/man/woman.

- **Gender role:** the set of expected behaviors, attitudes, and traits for one's gender (“boys don't cry”).
- **Gender identity:** one's personal sense of being male, female, or a combination.
- **Gender typing:** the process of acquiring a traditional masculine or feminine role.

Sex and gender both shape socialization — children absorb norms from parents, peers, media, and culture long before they can articulate them.

Sexual orientation: one's attraction — the thoughts, feelings, and actions related to physical attraction to another person. Research in this area is typically **NON-experimental** (surveys, correlational designs), because sexual orientation is not a manipulable variable; that limits causal conclusions.

Misconception: gender identity, gender role, and sexual orientation are distinct constructs — knowing one does not determine the others.

Exam tip: scenario language maps directly — “expects to be a boy” = gender identity; “plays only with trucks because ‘boys do’” = gender typing; “who you are attracted to” = sexual orientation.

Piaget's Cognitive Development

id: dev-piaget · Unit 3 — Development and Learning · core piaget schemas conservation cognitive-development

Piaget's core idea: children are little scientists who actively construct schemas — mental frameworks — through **assimilation** (fitting new information into existing schemas) and **accommodation** (modifying schemas when new information does not fit).

The four stages:

1. **Sensorimotor (birth to about 2):** knowing the world through sensory impressions and motor actions; **object permanence** — things exist when unseen — develops here.
2. **Preoperational (about 2-7):** language and mental symbols emerge; **pretend play** flourishes; BUT **egocentrism** (cannot take another's viewpoint), **animism** (believing objects are alive), and **lack of conservation** (quantity is not understood as invariant across shape changes) and lack of reversibility; **theory of mind** begins developing.

3. **Concrete operational (about 7-11):** logical, realistic thinking about CONCRETE events; **conservation mastered**; reversibility gained; still struggle with abstraction.
4. **Formal operational (about 12+):** abstract and hypothetical reasoning; systematic logic (Piaget acknowledged not everyone fully achieves this stage).

Theory of mind: understanding that others have their own minds, beliefs, and perspectives — tested with false-belief tasks; develops through the preoperational years.

Misconception: conservation failure is about APPEARANCE — the child judges by how it LOOKS (taller glass = “more water”), not by logic; and egocentrism is not selfishness — the child simply cannot yet model another’s perspective.

Exam tip: stage questions hinge on the missing ability: no object permanence = sensorimotor; no conservation = preoperational; logical but concrete = concrete operational; hypotheticals OK = formal operational.

Vygotsky's Sociocultural Theory & Cognitive Aging

id: dev-vygotsky-aging · Unit 3 — Development and Learning · [standard](#) [vygotsky](#) [zpd](#) [scaffolding](#) [aging](#)

Vygotsky’s sociocultural theory: children learn SOCIALLY, through interaction within cultural contexts — learning is embedded in sociocultural context.

- **Scaffolding:** temporary support (from a teacher, parent, or more expert peer) that helps a learner accomplish a task, gradually withdrawn as competence grows.
- **Zone of proximal development (ZPD):** the gap between what a child can do ALONE and what they can do WITH HELP — the ideal learning zone. Instruction aimed inside the ZPD produces the most growth; what a child can already do alone adds little, and what they cannot do even with help is out of reach for now.

Piaget vs. Vygotsky: Piaget placed the child exploring alone at the center; Vygotsky placed the SOCIAL and CULTURAL interaction at the center — the same child, different engines of development.

Aging and cognition: fluid intelligence (speedy abstract reasoning) wanes in late adulthood, while crystallized intelligence (accumulated knowledge and vocabulary) remains stable or grows. **Dementia** — cognitive disorders of adulthood impairing memory and thinking (e.g., Alzheimer’s disease) — is pathology, not normal aging.

Misconception: scaffolding is not doing the task for the child — support is calibrated and temporary, always aimed at transferring independence.

Exam tip: ZPD scenarios describe what a learner does WITH assistance — the exam answer names the ZPD and the scaffolding (hints, modeling, prompts) rather than simply “teaching.”

Communication & Language Development

id: dev-language · Unit 3 — Development and Learning · [core](#) [language](#) [development](#) [critical-period](#) [whorf](#)

Language: a shared, rule-governed, generative system of arbitrary symbols.

- **Phonemes:** the smallest distinctive SOUND units (“that” = th, a, t).
- **Morphemes:** the smallest units CARRYING MEANING (“readers” = read + er + s).
- **Semantics:** the rules for deriving meaning.
- **Grammar:** the overall system of rules enabling communication.
- **Syntax:** the rules for combining words into sentences (word order).

Language development stages (universal order):

1. **Cooing** — about 4 months (“ah-goo”).
2. **Babbling** — about 10 months; babbling comes to resemble the household language (“ma-ma”).
3. **One-word stage** — about 12 months (“Kitty!”).

4. **Telegraphic speech** — about 24 months; mostly nouns plus verbs, like a telegram (“Get ball”).
5. Rapid development into full sentences (24+ months).

Overgeneralization: applying grammar rules too broadly (“goed” instead of “went”) — paradoxical EVIDENCE of rule-learning.

Language and the brain: Broca’s area (left frontal — expression; damage gives effortful, broken speech); **Wernicke’s area** (left temporal — comprehension; damage gives fluent nonsense); **aphasia** = language impairment from left-hemisphere damage. Broca-impaired patients may still SING familiar songs.

Innateness and the critical period: humans show a predisposition for language (the universal grammar concept); without language exposure by about age 7, full acquisition may be permanently lost.

Language and thought (Whorf):

- **Linguistic determinism (strong version):** language DETERMINES — strictly shapes and limits — thought; one cannot think about concepts for which one’s language has no words.
- **Linguistic relativism (weak version, linguistic relativity):** language merely INFLUENCES thought — speakers of different languages attend to and categorize the world somewhat differently (a language with many color words draws attention to color distinctions).

Misconception: babbling is not “practicing words” — early babbling contains sounds from ALL languages; only later does it narrow to the household language. And overgeneralization errors show the child LEARNED the rule (add -ed for past), not that they are regressing.

Exam tip: determinism vs. relativism is the classic discrimination — determinism says language LIMITS thought outright; relativism says it BIASES thought. The exam treats the relativist (weak) view as the defensible one.

Ecological Systems & Parenting Styles

id: dev-ecological-parenting · Unit 3 — Development and Learning · **standard** bronfenbrenner parenting development

Bronfenbrenner’s ecological systems theory: development is shaped by nested environmental systems —

- **Microsystem:** direct-contact groups and environments (family, school).
- **Mesosystem:** interactions BETWEEN microsystems (parent-teacher relationships).
- **Exosystem:** indirect influences (a parent’s workplace).
- **Macrosystem:** cultural values, laws, customs, norms.
- **Chronosystem:** the time dimension — life transitions and historical events.

Parenting styles (demandingness x responsiveness):

- **Authoritarian** — high demandingness (rules, obedience: “Because I said so”), LOW responsiveness. Children: lower self-esteem, weaker social skills, overreactive brain response to mistakes.
- **Authoritative** — high demandingness PLUS high responsiveness (rules with open discussion and flexibility). Children: higher self-esteem, self-reliance, self-regulation, social competence.
- **Permissive** — low demandingness, high responsiveness. Children: greater aggression, immaturity.
- **Negligent (uninvolved)** — low on both. Children: poor academic and social outcomes.

Cultural differences exist in how these styles play out — the authoritative advantage is broadly found but expressed differently across cultures.

Mnemonic: think “demanding x responsive” as a 2x2 grid; authoritative is high on BOTH (the “best of both”).

Misconception: authoritarian is not the “strict-but-loving” style — strict with warmth is AUTHORITATIVE; authoritarian is strict WITHOUT warmth.

Exam tip: match outcomes to styles from the scenario's tone — rules plus dialogue = authoritative; “because I said so” = authoritarian; warm but no limits = permissive; absent = negligent.

Attachment & Temperament

id: dev-attachment · Unit 3 — Development and Learning · **core** attachment harlow temperament development

Attachment: the emotional tie to a caregiver, shown by seeking closeness.

Attachment styles (strange-situation patterns):

- **Secure attachment:** comfortable with the mother present; distressed when she leaves; comforted when she returns.
- **Insecure-avoidant:** indifferent to departure and return.
- **Insecure-anxious/resistant:** extreme separation distress; not easily comforted upon return; clinging.
- **Insecure-disorganized:** no consistent pattern; clingy and distressed; linked with abuse or neglect.

Contact comfort (Harlow's monkeys): infant monkeys preferred the cloth “mother” over the wire “mother” that fed them — COMFORT/CONTACT, not food, is the basis of attachment.

- **Stranger anxiety:** fear of strangers beginning around 8 months.
- **Separation anxiety:** heightened distress when separated from the caregiver.
- **Temperament:** characteristic emotional reactivity and intensity — stable from infancy and related to attachment style (easygoing vs. difficult babies).

Misconception: Harlow's experiment is about CONTACT comfort, not “love of food” — it refuted the drive-reduction feeding account of attachment; and insecure attachment is not “no attachment” — it is a disorganized or avoidant STRATEGY.

Exam tip: scenario cue “cries when mother leaves, calms quickly when she returns” = secure; “ignores her exit and return” = avoidant; “clings yet resists comfort” = anxious/resistant.

Peer Relationships & Adolescent Development

id: dev-peers-adolescence · Unit 3 — Development and Learning · standard peers adolescence egocentrism play

Peer relationships in childhood:

- **Parallel play:** playing ALONGSIDE but not directly WITH peers (typical of toddlers).
- **Pretend play:** symbolic, imaginative shared play (preoperational stage) — a forum for theory-of-mind practice.
- Adolescents gradually rely more on PEERS than family for identity and support.

Adolescent egocentrism (two faces):

- **Imaginary audience:** the belief that everyone is watching and judging you (a pimple feels like a headline).
- **Personal fable:** the belief that one is unique, special, and invulnerable (“it can't happen to me” — the logic of risky behavior).

Brain development: the **prefrontal cortex** — impulse control and decision-making regions — matures into the MID-20s, while emotional limbic systems mature earlier. This imbalance explains teen impulsivity and emotionality: full accelerator, unfinished brakes.

Adulthood timing: the **social clock** — culturally preferred timing of life events (marriage, parenthood, retirement). **Emerging adulthood:** in Western cultures, roughly age 18 to the mid-20s — no longer adolescents but not yet fully independent adults.

Misconception: adolescent risk-taking is not “stupidity” — knowledge is intact; the developing prefrontal control systems plus heightened reward sensitivity and the personal fable drive it.

Exam tip: imaginary audience = “everyone is LOOKING at me”; personal fable = “nothing bad can happen to ME” — match the belief to its face of adolescent egocentrism.

Identity, Erikson's Stages & Social Identity

id: dev-identity-erikson · Unit 3 — Development and Learning · core erikson identity marcia self-awareness

Erikson’s psychosocial stages — eight crises across the lifespan, each a turning point:

1. **Infancy (0-1): Trust vs. Mistrust** — reliable needs produce basic trust.
2. **Toddlerhood (1-3): Autonomy vs. Shame and Doubt** — exercising will and independence.
3. **Preschool (3-6): Initiative vs. Guilt** — initiating tasks and plans.
4. **Elementary (6-puberty): Industry vs. Inferiority** — competence through effort.
5. **Adolescence (teens-20s): Identity vs. Role Confusion** — refining a sense of self by testing roles.
6. **Young adulthood (20s-40s): Intimacy vs. Isolation** — forming close relationships, capacity for love.
7. **Middle adulthood (40s-60s): Generativity vs. Stagnation** — contributing to the next generation and the world.
8. **Late adulthood (60s+): Integrity vs. Despair** — life reflection ends in satisfaction or regret.

(Freud’s psychosexual stages are outside the CED scope.)

Identity development (Marcia’s statuses — taught):

- **Achievement:** explored options AND committed.
- **Foreclosure:** committed WITHOUT exploring (adopted parents’ or culture’s path).
- **Moratorium:** exploring, not yet committed.
- **Diffusion:** neither exploring nor committed.

Identity domains: racial/ethnic identity, gender identity, sexual orientation, religious identity, occupational identity, familial identity — developed partly through considering **possible selves** (who one might become).

Self-awareness: recognition of oneself as a SEPARATE INDIVIDUAL with one’s own thoughts, feelings, and traits — emerges in infancy/toddlerhood (classic test: mirror self-recognition, 18-24 months) and underlies later self-concept, self-esteem, and identity.

Social identity: the part of a person’s self-concept DERIVED FROM GROUP MEMBERSHIPS (race/ethnicity, gender, religion, nationality, teams, clubs) — “who I am” as defined by the groups I belong to; it feeds in-group identity and bias (see Unit 4).

Misconception: a crisis is not a breakdown — in Erikson each stage is a developmental TASK; and foreclosure sounds like stability but skips the exploration that real identity achievement requires.

Exam tip: Marcia questions hinge on the two levers — exploration and commitment. Ask of the scenario: did they explore? did they commit?

Classical Conditioning

id: dev-classical-conditioning · Unit 3 — Development and Learning · core classical-conditioning pavlov learning

association

Behaviorism: the view that psychology should be the objective science of observable behavior, without reference to mental processes. **Associative learning:** learning that certain events occur together — two stimuli (classical) or behavior-consequence (operant).

Classical conditioning (Pavlov): learning to link two or more stimuli and anticipate events.

- **UCS (unconditioned stimulus):** naturally, automatically triggers a response (food).
- **UCR (unconditioned response):** the unlearned, natural response to the UCS (salivation).
- **CS (conditioned stimulus):** an originally neutral stimulus that, after association with the UCS, triggers a response (bell).
- **CR (conditioned response):** the LEARNED response to the CS (salivation to the bell).

Process concepts:

- **Acquisition:** the initial stage when the association is learned; the CS-UCS ORDER matters (the CS must predict the UCS).
- **Extinction:** the CR diminishes when the CS is repeatedly presented WITHOUT the UCS.
- **Spontaneous recovery:** reappearance of an extinguished CR after a rest pause (weakened).
- **Stimulus generalization:** the CR is triggered by stimuli SIMILAR to the CS (Little Albert's fear spread to other furry objects).
- **Stimulus discrimination:** the learned ability to distinguish the CS from irrelevant stimuli — responding only to the specific CS.
- **Higher-order conditioning:** using an established CS as a “UCS” to condition a NEW neutral stimulus (bell, then light, then salivation to light).
- **Counterconditioning:** the therapeutic use of classical conditioning — replacing a CR (fear) with an incompatible response (relaxation); the basis of many treatments.
- **Conditioned emotional responses:** emotions (fear) can be classically conditioned (Little Albert; phobias).
- **Taste aversion:** learned avoidance of a food after one bad experience.
- **One-trial conditioning:** the association forms after a SINGLE pairing and is not strengthened by further pairings — adaptive for survival.
- **Biological preparedness:** organisms are biologically predisposed to learn CERTAIN associations more easily (taste-illness over tone-illness).
- **Habituation:** a diminished response to a repeated, non-harmful stimulus; in infants, waning interest indicates familiarity learning.

Mnemonic: UCS/UCR are UNlearned (nature); CS/CR are learned (conditioned). The CR and UCR look similar but the CR answers to the CS.

Misconception: classical conditioning is involuntary (reflexes, emotions, bodily responses) — voluntary behaviors belong to operant conditioning; and in taste aversion, the “CS” can work even hours after the UCS because preparedness overrides normal contiguity rules.

Exam tip: identify the four elements in order — the thing that ALREADY caused the response is the UCS; the thing that only caused it AFTER pairing is the CS.

Operant Conditioning

id: dev-operant-conditioning · Unit 3 — Development and Learning · **core** operant-conditioning reinforcement
punishment skinner

Operant conditioning (Skinner): learning in which behavior is strengthened by reinforcers or diminished by punishers — CONSEQUENCES shape voluntary behavior. **Law of Effect (Thorndike):** behaviors followed by favorable consequences become more likely; behaviors followed by unfavorable consequences become less likely. The **Skinner box** is the chamber with a bar or key animals manipulate for reinforcers, with recording devices.

The four consequences (positive = ADD a stimulus; negative = REMOVE a stimulus):

- **Positive reinforcement:** ADD something pleasant to INCREASE behavior (praise, treats).
- **Negative reinforcement:** REMOVE something unpleasant to INCREASE behavior (buckling the seatbelt stops the beeping).
- **Positive punishment:** ADD something unpleasant to DECREASE behavior (scolding, a ticket).

- **Negative punishment:** REMOVE something pleasant to DECREASE behavior (taking away the phone).

Reinforcers: **primary** — innately satisfying, fulfills a biological need (food, water, warmth); **secondary/conditioned** — gains power through association with a primary reinforcer (money, grades, praise); **immediate** — given right after the behavior (more powerful); **delayed** — given after a delay (paycheck, exam results; requires self-control).

Key processes: **shaping** — reinforcing successive approximations of the desired behavior; **reinforcement discrimination** — responding only when specific cues predict reinforcement; **reinforcement generalization** — responding similarly to related cues; **instinctive drift** — learned behaviors drift back toward biologically predisposed patterns (raccoons “washing” tokens); **superstitious behavior** — repeating behaviors mistakenly associated with reinforcement; **learned helplessness (Seligman’s dogs)** — when unable to avoid repeated aversive events, organisms learn they have no control and passively give up, even when escape later becomes possible.

Reinforcement schedules:

- **Continuous reinforcement:** every correct response reinforced — fast acquisition, fast extinction.
- **Partial (intermittent) reinforcement:** only sometimes reinforced — slower acquisition, much GREATER resistance to extinction.
 - **Fixed-ratio (FR):** after a SET NUMBER of responses (every 5th press); high, steady responding.
 - **Variable-ratio (VR):** after an UNPREDICTABLE number of responses (gambling, slot machines); highest, most persistent response rate.
 - **Fixed-interval (FI):** first response after a FIXED TIME (weekly paycheck); produces a SCALLOPED pattern — a surge near reinforcement time, a drop after.
 - **Variable-interval (VI):** first response after an UNPREDICTABLE time (pop quizzes, fishing); slow, steady responding.

Misconception: negative reinforcement is NOT punishment — both “negative” words mean removal, but reinforcement (either kind) INCREASES behavior while punishment DECREASES it. And the partial-reinforcement effect explains why gambling habits are so durable.

Exam tip: two questions decode any consequence — did behavior go UP (reinforcement) or DOWN (punishment)? was something ADDED (positive) or REMOVED (negative)?

Social, Cognitive & Neurological Factors in Learning

id: dev-social-cognitive-learning · Unit 3 — Development and Learning · **core** observational-learning bandura
latent-learning mirror-neurons

Social learning theory (Bandura): we learn by observing and imitating others (models) — no direct experience required.

- **Observational learning:** learning by watching models; the more SIMILAR the model, the more likely the imitation.
- **Modeling:** the process of observing and imitating a specific behavior.
- **Bobo doll experiment (Bandura):** children who watched adults abuse the Bobo doll imitated the aggression — demonstrating observational learning of aggression.
- **Vicarious conditioning:** learning consequences by watching OTHERS being rewarded or punished (you need not touch the hot stove yourself).

Cognitive learning (learning without reinforcement):

- **Insight learning (Kohler’s chimpanzees):** the sudden realization of a solution without trial-and-error — the “aha.”
- **Latent learning (Tolman’s rats):** learning that occurs WITHOUT reinforcement and is not demonstrated until an incentive appears (rats formed maps of mazes but only raced when food appeared).

- **Cognitive maps:** mental representations of an environment's layout.

Neurological factor: mirror neurons — neurons that fire both when performing an action and when OBSERVING someone else perform it (basis for imitation and empathy; demonstrated in monkeys). (Flagged beyond the CED's required terms, but the concept underpins observational learning.)

Misconception: latent learning shows learning HAPPENED without showing in PERFORMANCE until incentives change — learning and performance are separable; and mirror neurons support imitation but “mirror” firing is not the same as understanding intention.

Exam tip: the classic application — a child swears after hearing a parent swear (observational learning), then is praised by friends for it (vicarious reinforcement); both Bandura concepts can appear in one scenario.

Unit 4 — Social Psychology and Personality

Attribution Theory & Person Perception

id: soc-attribution-perception · Unit 4 — Social Psychology and Personality

core

attribution

bias

person-perception

social-cognition

Attribution theory: how people explain the causes of behavior — their own and others'.

- **Dispositional (internal) attribution:** behavior explained by personality, skills, talent, IQ.
- **Situational (external) attribution:** behavior explained by circumstances beyond the person's control.
- **Explanatory style:** one's habitual way of explaining good and bad events — **optimistic** (bad events are external, unstable, specific) vs. **pessimistic** (bad events are internal, stable, global); pessimism links to depression.

Attribution biases:

- **Fundamental attribution error (FAE):** explaining OTHERS' behavior by overestimating personality and underestimating situational forces (the driver who cuts you off is “a jerk,” not “rushing to the hospital”).
- **Actor/observer bias:** we attribute OUR behavior to situations but OTHERS' behavior to dispositions.
- **Self-serving bias:** attributing our SUCCESSES to internal factors and our FAILURES to external factors.

Locus of control:

- **Internal:** the belief that one controls one's own outcomes (linked to achievement and well-being).
- **External:** the belief that outcomes are controlled by luck, fate, or others.

Person perception:

- **Mere exposure effect:** repeated exposure to a stimulus increases liking of it.
- **Self-fulfilling prophecy:** a belief or expectation that elicits behavior confirming itself (a teacher expects success, treats the student differently, the student succeeds).
- **Social comparison:** evaluating oneself by comparison with others — **upward comparison** (to those better off; can inspire or discourage) and **downward comparison** (to those worse off; boosts self-esteem).
- **Relative deprivation:** perceiving oneself as worse off relative to those with whom one compares.

Misconception: the FAE is about OTHERS' behavior; self-serving bias is about OURS. And actor/observer is not a contradiction of the FAE — it is the same asymmetry applied to ourselves.

Exam tip: scenario cue “she blamed the traffic, but he’s just lazy” = actor/observer bias in one sentence; “our win shows skill, our loss shows the refs” = self-serving bias.

Attitude Formation, Stereotypes & Prejudice

id: soc-attitudes-prejudice · Unit 4 — Social Psychology and Personality · core attitudes prejudice stereotypes

implicit-bias

Attitude: beliefs and feelings that predispose responses to objects, people, and events.

The prejudice triad (know the differences):

- **Stereotype:** a generalized (sometimes accurate, often overgeneralized) BELIEF about a group; it reduces cognitive load but frequently underlies bias.
- **Prejudice:** an unjustifiable (usually negative) ATTITUDE toward a group and its members.
- **Discrimination:** unjustifiable negative BEHAVIOR toward a group and its members.
- **Implicit attitudes:** unconscious beliefs and feelings that influence behavior — measured by the Implicit Association Test (IAT).

Biases underlying prejudice:

- **Just-world phenomenon:** the belief that people get what they deserve and deserve what they get (blames victims).
- **In-group bias:** favoring one’s own group; perceiving in-group members as diverse.
- **Out-group homogeneity bias:** perceiving out-group members as all alike (“they all look the same”).
- **Ethnocentrism:** the belief in the superiority of one’s own ethnic or cultural group.
- **Scapegoating:** blaming an out-group to discharge frustration (taught, though flagged beyond the CED’s required list).
- **Other-race effect:** recognizing own-race faces more accurately (taught).

Attitude change:

- **Belief perseverance:** clinging to initial beliefs even after receiving contradictory evidence — often fueled by confirmation bias.
- **Cognitive dissonance (Festinger):** see the persuasion entry.

Mnemonic: belief (stereotype), feeling (prejudice), action (discrimination) — “B-F-A.”

Misconception: prejudice is an ATTITUDE and can exist without discrimination (behavior); a person may hold a stereotype without acting on it — but implicit attitudes can leak into behavior unnoticed.

Exam tip: the IAT measures IMPLICIT attitudes — speeded pairings of concepts, not self-report; scenarios where someone sincerely denies prejudice yet shows biased reaction times are IAT items.

Cognitive Dissonance, Persuasion & Compliance

id: soc-dissonance-persuasion · Unit 4 — Social Psychology and Personality · standard dissonance persuasion

elm compliance

Cognitive dissonance (Festinger): the discomfort felt when actions and attitudes conflict; we reduce it by changing BEHAVIOR or changing ATTITUDES. Classic evidence: after choosing between two similarly attractive items (the toaster vs. microwave study), people rate their chosen item HIGHER — the choice created dissonance, resolved by inflating the chosen option.

Persuasion — the Elaboration Likelihood Model (two routes):

- **Central route:** persuasion by logic, evidence, and argument quality — requires thoughtful processing; produces deeper, more durable attitude change.

- **Peripheral route:** persuasion by superficial cues — attractiveness, credibility, emotion, the sheer number of arguments. The **halo effect** operates here: one positive trait (like beauty) colors overall judgment of a person or product.

Compliance techniques:

- **Foot-in-the-door:** a SMALL request first makes people more likely to agree to a LARGER later request (consistency pressure — “I’m the kind of person who helps”).
- **Door-in-the-face:** a LARGE unreasonable request first makes the smaller (real) request more likely accepted (reciprocal concession).

Misconception: dissonance is not simple regret — it specifically follows a FREE CHOICE or effortful act that conflicts with self-image; and the central route requires motivation AND ability to think, not just good arguments.

Exam tip: route questions hinge on WHAT persuaded: evidence digested thoughtfully = central; celebrity endorsement = peripheral. Foot-in-door escalates a small YES; door-in-face retreats from a big NO.

Conformity, Obedience & Culture

id: soc-conformity-obedience · Unit 4 — Social Psychology and Personality · **core** conformity obedience
social-influence culture

Social norms: unwritten rules and expectations governing acceptable behavior.

Social influence theory — two reasons we conform:

- **Normative social influence:** conforming to gain approval or avoid rejection (laughing at a joke you did not get).
- **Informational social influence:** conforming because we believe others’ interpretation is more accurate (looking up when everyone else does).

Conformity: adjusting behavior or thinking to match a group standard (Asch’s line-judgment studies — people denied clear perception to agree with a unanimous wrong majority). **(The named studies are studied conceptually, though outside the strict CED list.)**

Obedience: following orders from an authority figure (Milgram’s shock experiments — about 65% obeyed to the maximum).

Misconception: conformity is matching PEERS or implicit group expectations; obedience is complying with an explicit COMMAND from an authority. If nobody ordered anything, it is not obedience.

Culture:

- **Individualism:** prioritizing personal goals and identity (“I”).
- **Collectivism:** prioritizing group harmony and identity (“we”).
- **Multiculturalism:** valuing multiple cultural identities within a society.

Exam tip: the classic confederate line-up — if confederates give wrong answers and the participant follows, that is conformity (normative influence); if an experimenter in a lab coat instructs increasing shocks, that is obedience.

Group Dynamics & the Bystander Effect

id: soc-group-dynamics · Unit 4 — Social Psychology and Personality · **core** groups groupthink bystander-effect
deindividuation

Group dynamics:

- **Group polarization:** group discussion STRENGTHENS the group’s prevailing (already leaning) opinion — groups end up more extreme than their members’ average starting view.

- **Groupthink:** the desire for harmony overrides realistic evaluation of alternatives — bad decisions because dissent is suppressed.
- **Social facilitation:** improved performance on WELL-LEARNED tasks in others' presence (impaired on difficult ones).
- **Social loafing:** exerting less effort in a group task than when alone (tug-of-war studies).
- **Deindividuation:** loss of self-awareness and restraint in group situations fostering anonymity and arousal (mobs, costumes).
- **Diffusion of responsibility:** feeling less personally responsible when others are present.

Helping:

- **Bystander effect:** individuals are LESS likely to help when others are present (Kitty Genovese case; Darley and Latane's seizure study) — caused by diffusion of responsibility plus **pluralistic ignorance** (everyone privately rejects the norm but assumes others accept it, so no one acts — taught, though flagged).
- **Altruism:** selfless concern for and helping of others (see the aggression/altruism entry).

Perceptual biases in groups (taught, though flagged): **false consensus effect** — overestimating how much others share our beliefs and behaviors; **spotlight effect** — overestimating how much others NOTICE us; **social inhibition** — performance impairment in others' presence on difficult tasks.

Misconception: social facilitation and social inhibition are two halves of the same law — presence of others energizes the DOMINANT response: right for easy/well-learned tasks (facilitation), wrong for hard/novel ones (inhibition).

Exam tip: the classic emergency scenario — more bystanders means LESS helping; the mechanism chain is diffusion of responsibility and pluralistic ignorance, not apathy.

Attraction & Relationships

id: soc-attraction · Unit 4 — Social Psychology and Personality · [standard](#) [attraction](#) [proximity](#) [similarity](#)
[self-disclosure](#)

Factors in interpersonal attraction:

- **Proximity:** geographic nearness — the most powerful predictor of relationship formation; it creates opportunity for repeated contact, and repeated exposure breeds liking (the **mere exposure effect**, Unit 4.1).
- **Physical attractiveness:** the strongest initial pull in first encounters; attractive people are unconsciously judged as kinder, healthier, and more successful (the **halo effect**, Unit 4.2).
- **Similarity:** shared attitudes, values, and interests sustain attraction — “birds of a feather flock together”; compatibility matters more than opposites.
- **Reciprocal liking:** we tend to like people who like US — believing someone likes us makes us act more likably toward them.
- **Self-disclosure:** gradually sharing intimate thoughts and feelings deepens closeness; MUTUAL self-disclosure builds liking and relationships.

How the factors fit together: proximity and attractiveness open the door; similarity and reciprocal liking keep the conversation going; self-disclosure deepens the relationship over time.

Misconception: “opposites attract” is not the tested rule — similarity is the robust finding; and proximity works through repeated EXPOSURE, not mere physical distance on a map.

Exam tip: scenario cues — “same dorm floor, kept running into each other” = proximity/mere exposure; “she laughed at his jokes after learning he liked her” = reciprocal liking; “late-night conversations about childhood” = self-disclosure.

Aggression, Altruism & Cooperation/Conflict

id: soc-aggression-altruism · Unit 4 — Social Psychology and Personality · **standard** aggression altruism conflict
social-scripts

Aggression: any physical or verbal behavior **INTENDED** to hurt or destroy. Influenced by biological factors (amygdala activity, testosterone, low serotonin, genetic predisposition), aversive situations (frustration, crowding, provocation), and learning (observing aggressive models — Bandura, Unit 3.9).

- **Frustration-aggression principle:** **FRUSTRATION** — the blocking of goal-directed behavior — creates anger, which can trigger aggression (displaced anger after a defeat).
- **Social scripts:** culturally provided mental guides for **HOW** to behave in familiar situations; media violence supplies **AGGRESSION SCRIPTS** that are rehearsed through observational learning.

Cooperation and conflict:

- **Superordinate goals:** shared goals requiring cooperation that override differences and reduce hostility between groups (the Robbers Cave truck that needed both teams to pull).
- **Social traps:** situations where individuals pursuing self-interest harm the collective (overfishing, pollution).
- **Industrial-organizational (I/O) psychology:** applies psychology to the workplace — management, teamwork, employee well-being (burnout).

Prosocial behavior:

- **Altruism:** selfless concern for and helping of others.
- **Social reciprocity norm:** we help those who have helped us.
- **Social responsibility norm:** we should help those who need help and depend on us.

Misconception: aggression requires **INTENT** — accidental harm is not aggression; and altruism is defined by the **HELPER**'s motive, not the size of the good deed.

Exam tip: script questions link media exposure to behavior **THROUGH** learning (rehearsed scripts), not through catharsis; superordinate-goal questions stress that only a **SHARED** goal neither group can reach alone reduces prejudice.

Psychodynamic Theory of Personality (Freud)

id: soc-psychodynamic · Unit 4 — Social Psychology and Personality · **core** freud psychodynamic
defense-mechanisms unconscious

Core claim: personality arises from **UNCONSCIOUS** processes — a reservoir of mostly unacceptable thoughts, wishes, and memories outside awareness.

Structure of personality:

- **Id:** the pleasure principle (immediate gratification); present from birth; the unconscious reservoir of basic sexual and aggressive drives.
- **Ego:** the reality principle (realistic satisfaction); emerges around ages 2-3; the largely conscious “executive” mediating among id, superego, and reality.
- **Superego:** morality and conscience; emerges around ages 3-5; internalized ideals of right and wrong from parents and society.

Ego defense mechanisms (unconscious processes protecting the ego from anxiety):

- **Denial:** refusing to accept reality (a crime victim insisting it never happened).
- **Displacement:** redirecting emotions to a safer target (angry at the boss, yells at the spouse).
- **Projection:** attributing one's own feelings to others (fearing your spouse cheats, you accuse them).
- **Rationalization:** logical-but-false justifications (denied a loan: “the house was too big anyway”).
- **Reaction formation:** acting opposite of true feelings (sad about a breakup, acting ecstatic).
- **Regression:** retreating to childlike behavior (hugging a teddy bear under stress).

- **Repression:** banishing distressing thoughts from consciousness (abused but not remembering).
- **Sublimation:** channeling impulses into acceptable outlets (upset with a spouse, going for a long run).
- **Intellectualization** (taught): detached, academic processing to avoid emotion (calmly making funeral arrangements).

Assessment — projective tests: ambiguous stimuli designed to probe the unconscious — the **Rorschach Inkblot Test** (inkblots) and the **Thematic Apperception Test (TAT)** (ambiguous pictures that elicit stories).

Mnemonic: the id WANTS (pleasure), the ego THINKS (reality), the superego JUDGES (morality).

Misconception: defense mechanisms are normal and unconscious — using one is not automatically pathological; they become a problem only when rigid and pervasive.

Exam tip: mechanism questions hinge on the transformation: same feeling, new TARGET (displacement); feeling reversed (reaction formation); feeling rebranded as logic (rationalization/intellectualization); impulse upgraded into something productive (sublimation).

Humanistic Theory of Personality

id: soc-humanistic · Unit 4 — Social Psychology and Personality · **standard** humanistic rogers self-actualization upr

Humanistic theories focus on human GROWTH POTENTIAL rather than on disorders — the “third force” after behaviorism and psychoanalysis.

- **Self-actualizing tendency:** the innate drive to fulfill one’s potential — the engine of personality in this view.
- **Unconditional positive regard (UPR):** total, non-judgmental acceptance of a person — needed for a healthy self-concept (Rogers). With conditional regard, children learn to suppress their true self to earn approval.
- **Self-concept:** how one views oneself in relation to others; the closer the match between the ideal self and the actual self, the greater the well-being.

(Maslow’s hierarchy of needs is taught for context, though outside the CED scope: physiological and safety needs, then belonging and esteem, with self-actualization at the peak.)

Misconception: unconditional positive regard does not mean approving of every behavior — the PERSON is accepted while behavior can still be limited; and humanistic theories are criticized as vague and hard to test, not as wrong.

Exam tip: therapy scenarios with a therapist who listens without judgment, reflects feelings, and trusts the client’s capacity to grow = humanistic/person-centered (Rogers) — connect to UPR explicitly for the point.

Social-Cognitive & Trait Theories of Personality

id: soc-social-cognitive-trait · Unit 4 — Social Psychology and Personality · **core** personality big-five reciprocal-determinism traits

Social-cognitive theory (Bandura): personality is shaped by the INTERACTING influences of **reciprocal determinism** — behavior, personal/cognitive factors, and environment each influence the others in a triangle.

- **Self-concept:** how one views oneself in relation to others.
- **Self-efficacy:** belief in one’s ability to SUCCEED AT SPECIFIC TASKS — predicts effort and persistence.
- **Self-esteem:** one’s general sense of self-WORTH.

Trait theory:

- **Trait:** a characteristic pattern of behavior or disposition — a stable disposition leading to typical responses.
- **The Big Five (OCEAN)** — the dominant trait model:
 - **Openness:** conventional/uncreative at the low end; imaginative, curious, original at the high end.
 - **Conscientiousness:** lazy, disorganized, negligent at the low end; hard-working, organized, punctual at the high end (the best predictor of school and job success).
 - **Extraversion:** loner, quiet, reserved at the low end; joiner, talkative, active at the high end.
 - **Agreeableness:** suspicious, critical, ruthless at the low end; trusting, soft-hearted, helpful at the high end.
 - **Neuroticism (vs. emotional stability):** calm, even-tempered at the low end; anxious, moody, self-conscious at the high end.
- **Personality inventories:** self-report questionnaires (e.g., Likert-scale ratings of statements) assessing traits.
- **Factor analysis:** the statistical method identifying clusters of related items — the basis of Big Five trait extraction.

(MMPI, NEO-PI, 16PF, and Type A/B are taught but not required by name.)

Mnemonic: OCEAN = Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism.

Misconception: self-esteem (worth) and self-efficacy (task competence) are different — a student can believe in their math ability (high efficacy) yet feel worthless overall (low esteem); traits describe tendencies, they do not enslave behavior (the person-situation debate is context).

Exam tip: reciprocal-determinism scenarios must show TWO-WAY influence — “a hostile child provokes harsh parenting, which deepens the hostility” — environment and person loop, and the essay answer should name all three corners.

Motivation & Hunger

id: soc-motivation · Unit 4 — Social Psychology and Personality · core motivation hunger drive-reduction self-determination

Motivation: a need or desire that energizes and directs behavior.

Theories:

1. **Drive-reduction theory:** a physiological need (hunger) creates an aroused DRIVE (hunger drive) that motivates behavior to satisfy the need and restore **homeostasis** (a balanced internal state). Sequence: need, drive, behavior, homeostasis.
2. **Arousal theory:** we seek an OPTIMAL level of arousal — not too little (bored) or too much (stressed). **Yerkes-Dodson law:** performance peaks at moderate arousal; difficult tasks need LOWER arousal, easy tasks tolerate HIGHER arousal.
3. **Incentive theory:** behavior is pulled by external rewards or incentives — positive or negative stimuli that attract or repel.
4. **Instinct theory:** innate, fixed patterns of behavior motivate actions in animals (salmon swimming upstream); humans show few clear instincts.
5. **Self-determination theory:** humans are motivated to fulfill three needs — **competence, autonomy, relatedness**. **Intrinsic motivation:** doing something for its inherent satisfaction; **extrinsic motivation:** doing it for external rewards or to avoid punishment. **Overjustification effect** (taught): extrinsic rewards can undermine intrinsic motivation (paying a child to draw, who already loves drawing).
6. **Lewin's motivational conflict theory:** choice creates conflicts — **approach-approach** (two desirable options, two colleges); **avoidance-avoidance** (two undesirable options, homework or

detention); **approach-avoidance** (one option with both attractive and unattractive features, tasty-but-fattening ice cream; produces vacillation).

7. **Sensation-seeking theory:** the need for varied, novel, intense experience — experience seeking, thrill/adventure seeking, disinhibition, boredom susceptibility.

Hunger motivation:

- **Stomach contractions** and **blood glucose** levels signal hunger.
- The **hypothalamus** (via the pituitary) regulates hunger and satiety: the **lateral hypothalamus** is the hunger “on” center; the **ventromedial hypothalamus** is the satiety “off” center.
- **Hormones that INCREASE appetite:** **ghrelin** (empty stomach: “I’m hungry”), **orexin** (hypothalamus hunger trigger). **Hormones that DECREASE appetite:** **insulin** (pancreas; controls blood glucose), **leptin** (fat cells; abundant leptin raises metabolism and reduces hunger), **PYY** (digestive tract: “I’m not hungry”).
- **Set point** (taught): the body’s regulated weight “thermostat”; glucose levels are monitored.
- **External cues:** the presence of food, time of day, and social gatherings also drive eating.
- **Affiliation need:** the need to build relationships and belong; **ostracism** (deliberate social exclusion) causes pain-like distress.
- **(nAch/nPow/nAff — achievement, power, affiliation motives — are taught, measured with TAT-type stories.)**

Misconception: negative reinforcement and drive reduction are different frames — drive reduction pushes from INSIDE (need); incentives PULL from outside; and leptin reducing hunger is “satiety,” not “more hunger” (a common reversal trap).

Exam tip: Yerkes-Dodson questions hinge on task difficulty — difficult tasks want LOWER arousal; easy tasks can use higher arousal. Motivation scenario “studied because the material is fascinating” = intrinsic.

Emotion

id: soc-emotion · Unit 4 — Social Psychology and Personality · **core** emotion theories facial-feedback

display-rules

Emotion (affect): a complex psychological process distinct from reasoning and knowledge; it reflects internal and external factors.

Theories of emotion — the CED tests the broad IDEAS, not the specific names (James-Lange, Cannon-Bard, Schachter two-factor, and Lazarus are excluded by name):

- Some theories: **physiological arousal FIRST, then emotion** — “I tremble, therefore I am afraid.”
- Some theories: **arousal and emotion occur SIMULTANEOUSLY.**
- Some theories: **a cognitive label or appraisal is required** — emotion equals arousal plus interpretation (“Is it dangerous?”); appraisal sometimes occurs without awareness.

Embodiment of emotion:

- **Facial-feedback hypothesis:** facial expressions INFLUENCE emotional experience (forcing a smile can mildly elevate mood); supports the “body before cognition” view, though research results are mixed.
- **Behavioral feedback:** body posture and gestures also influence emotions.

Broaden-and-build theory: positive emotions BROADEN awareness and encourage exploratory thoughts and actions (building resources); negative emotions narrow thinking to a fight-or-flight focus.

Universality of emotional expression: some emotions are expressed and recognized across cultures — commonly cited: **anger, disgust, sadness, happiness, surprise, fear** — though research shows mixed results.

Display rules: cultural, gender, age, and class norms regulating HOW and WHEN emotions may be expressed (some cultures discourage public grief; gender expectations about crying).

Misconception: the universality claim is about RECOGNIZING expressions, not about feeling identical emotions everywhere; display rules explain why the same emotion is shown differently across cultures.

Exam tip: you can answer theory questions by POSITION, not name — “heart races, THEN fear” = arousal-first; “heart races and fear together” = simultaneous; “arousal interpreted as fear in the dark alley” = cognitive-label. Facial-feedback items give a manipulated expression with a measured mood.

Unit 5 — Mental and Physical Health

Health Psychology & Stress

id: hp-health-stress · Unit 5 — Mental and Physical Health · core stress coping gas health

Health psychology: explores psychological factors in physical health, illness, and wellness.

Stress: the process of appraising and responding to events (stressors) that we perceive as threatening or challenging. Stress is linked to immune suppression, hypertension, headaches, and heightened disease susceptibility.

Stressors:

- **Eustress:** motivating, positive stress (the week before a well-prepared competition).
- **Distress:** debilitating, negative stress.
- Sources: catastrophes, major life changes, and **daily hassles** — small irritations that build up.
- **Adverse childhood experiences (ACEs):** abuse, neglect, and household dysfunction in childhood — stress sources with lifespan-long effects on relationships and health (with sociocultural differences in what counts as an ACE).

General Adaptation Syndrome (GAS, Selye) — the body’s three-phase stress response:

1. **Alarm reaction:** the sympathetic nervous system mobilizes — fight-flight-or-freeze.
2. **Resistance:** the body copes with the stressor; resources are drawn down while temperature, blood pressure, and arousal remain high.
3. **Exhaustion:** resources are depleted; greatest vulnerability to illness and collapse.

Tend-and-befriend theory: under stress, some people (mostly women, likely via oxytocin) respond by nurturing (tending) and seeking social connection (befriending) — a counterpoint to fight-or-flight.

Coping:

- **Problem-focused coping:** directly addressing the stressor — viewing stress as a problem to be solved; best when we HAVE CONTROL over the stressor.
- **Emotion-focused coping:** managing our emotional REACTIONS to stress — deep breathing, meditation, medication, seeking comfort; best when we CANNOT change the stressor.

Misconception: GAS phases are about the BODY’s resources, not mood — the danger of the resistance phase is that it feels like coping while the physiological cost accumulates toward exhaustion.

Exam tip: coping-style questions hinge on CONTROL: controllable stressor = problem-focused; uncontrollable = emotion-focused. Naming the wrong match is the classic lost point.

Positive Psychology

id: hp-positive-psychology · Unit 5 — Mental and Physical Health · standard well-being gratitude resilience
positive-psychology

Positive psychology: the scientific study of human flourishing — the factors promoting **well-being**, **resilience**, **positive emotions**, and **psychological health**, rather than focusing only on disorder.

- **Gratitude:** expressing gratitude increases **subjective well-being** (happiness) — gratitude journals and counted blessings reliably shift mood upward.
- **Signature strengths and virtues:** exercising one's core character strengths raises happiness. Six virtue categories: **wisdom, courage, humanity, justice, temperance, transcendence**.
- **Posttraumatic growth:** positive psychological change can follow struggle with highly challenging or traumatic life circumstances — deeper relationships, new possibilities, personal strength, changed priorities (distinct from the **DISTRESS** of posttraumatic stress, and the two can coexist).

Misconception: positive psychology is not “positive thinking” or the denial of problems — it is the empirical study of what makes life worth living, tested with the same methods as clinical research; and posttraumatic growth is not automatic — it emerges from the **STRUGGLE**, not from the event alone.

Exam tip: application items contrast flourishing with treating illness — gratitude intervention raising well-being scores, or growth after adversity, should be named as positive-psychology findings.

Defining & Classifying Psychological Disorders

id: hp-defining-disorders

Unit 5 — Mental and Physical Health

core

disorders

dsm

diagnosis

diathesis-stress

Defining disorder — three criteria considered TOGETHER (no single one suffices):

1. **Dysfunction:** the behavior impairs daily functioning.
2. **Distress:** the person experiences suffering.
3. **Deviation from social norms:** the behavior violates cultural expectations.

Consequences of diagnosis: positive — access to treatment, understanding, research; negative — **stigma**, racism, sexism, ageism, discrimination, and labeling effects.

Classification systems:

- **DSM (Diagnostic and Statistical Manual of Mental Disorders):** published by the American Psychiatric Association.
- **ICD (International Classification of Diseases):** published by the World Health Organization.
- Both are updated regularly with research advances; diagnosis requires specialized training.

Perspectives on etiology (causes):

- **Behavioral:** maladaptive learned associations between stimuli and responses.
- **Psychodynamic:** unconscious conflicts, often rooted in childhood.
- **Humanistic:** lack of social support; failure to fulfill potential.
- **Cognitive:** maladaptive thoughts, beliefs, attitudes, emotions.
- **Evolutionary:** behaviors that reduce survival or reproductive success.
- **Sociocultural:** maladaptive social or cultural relationships and dynamics.
- **Biological:** physiological and genetic issues.

Interaction models:

- **Biopsychosocial model:** disorders arise from interacting biological, psychological, and sociocultural factors.
- **Diathesis-stress model:** a genetic **VULNERABILITY** (diathesis) plus a stressful life experience produces the disorder — predisposition loads the gun; stress pulls the trigger.

Eclectic approach: most clinicians combine perspectives in diagnosis and treatment.

Misconception: deviance alone is not disorder (unusual but harmless behavior is not pathological), and distress alone is not always disorder either — the three criteria are weighed together.

Exam tip: diathesis-stress scenarios name **BOTH** the inherited vulnerability **AND** the stressful event — an answer crediting only one misses the model.

Neurodevelopmental & Schizophrenic Spectrum Disorders

id: hp-neurodevelopmental-schizophrenia

Unit 5 — Mental and Physical Health

standard

adhd

asd

schizophrenia

disorders

Neurodevelopmental disorders — onset during the developmental period; symptoms concern age-appropriateness. Causes: environmental, physiological, or genetic.

- **ADHD (attention-deficit/hyperactivity disorder):** inattention, hyperactivity, and impulsivity.
- **ASD (autism spectrum disorder):** deficits in social communication plus restricted, repetitive behaviors — a SPECTRUM, so severity and support needs vary widely.

Schizophrenic spectrum disorders: characterized by disintegration of logical thought processes; chronic or acute.

Positive symptoms (the PRESENCE of abnormal behaviors):

- **Delusions:** false beliefs — **paranoid** (persecution) and **grandiose** (grandeur) types are the listed examples.
- **Hallucinations:** false sensory perceptions — especially auditory (hearing voices).
- **Disorganized thinking/speech:** “word salad” — nonsensical word strings.
- **Catatonic excitement:** disordered, hyper motor behavior.

Negative symptoms (the ABSENCE of normal behaviors):

- **Flat affect:** absence of emotional expression.
- **Catatonic stupor:** unresponsiveness and lack of movement.

Possible causes: genetic and biological links — **prenatal virus exposure** and the **dopamine hypothesis** (excess dopamine activity).

Misconception: positive vs. negative does not mean good vs. bad — positive symptoms ADD abnormal behavior; negative symptoms REMOVE normal behavior. And schizophrenia is not “split personality” — it is a thought disorder; dissociative identity disorder is the “split” one.

Exam tip: classify each symptom first — hearing voices = positive (hallucination); showing no emotion = negative (flat affect). Medication questions connect to the dopamine hypothesis (antipsychotics block dopamine).

Depressive, Bipolar, Anxiety & OCD-Related Disorders

id: hp-mood-anxiety

Unit 5 — Mental and Physical Health

core

depression

bipolar

anxiety

ocd

Depressive disorders — sad, empty, or irritable mood plus physical and cognitive changes impairing functioning. Causes: biological, genetic, social, cultural, behavioral, and cognitive factors.

- **Major depressive disorder (MDD):** episodic severe depression.
- **Persistent depressive disorder:** chronic, lower-grade, longer-lasting depression.

Bipolar disorders — alternating periods of **mania** (elation, hyperactivity) and depression:

- **Bipolar I:** full manic episodes.
- **Bipolar II: hypomanic** episodes (milder mania) plus depression.

Anxiety disorders — excessive fear or anxiety with behavioral disturbance. Causes: learned associations, maladaptive cognition, biological and genetic factors.

- **Specific phobia:** fear of a specific object or situation (**acrophobia** = heights, **arachnophobia** = spiders).
- **Agoraphobia:** fear of situations difficult to escape — public transportation, open or enclosed spaces, crowds, being outside alone.

- **Panic disorder:** recurrent unexpected **panic attacks** — overwhelming biological, cognitive, and emotional surges of fear. Culture-bound pattern: **ataque de nervios** (Caribbean and Iberian descent — screaming, fainting, loss of control after distressing news).
- **Social anxiety disorder:** intense fear of being judged or watched by others. Culture-bound pattern: **taijin kyofusho** (Japan — fear that one's body offends or disgusts others).
- **GAD (generalized anxiety disorder):** prolonged, nonspecific anxiety and worry about many things.

Obsessive-compulsive and related disorders:

- **OCD: obsessions** (intrusive, unwanted thoughts) plus **compulsions** (repetitive behaviors that neutralize the obsessions). Distinct from obsessive-compulsive PERSONALITY disorder.
- **Hoarding disorder:** persistent difficulty discarding possessions.

Misconception: fear is tied to a PRESENT, specific threat; anxiety is more future-oriented and diffuse — GAD is worry about everything, not one thing. And panic attacks in panic disorder are UNEXPECTED; fear of the next attack itself drives the disorder.

Exam tip: MDD vs. persistent depressive = severe episodes vs. chronic low grade; bipolar I vs. II = full mania vs. hypomania. Culture-bound syndromes show that disorder EXPRESSION varies by culture while diagnostic thinking stays universal.

Dissociative, Trauma-Related & Eating Disorders

id: hp-dissociative-trauma-eating · Unit 5 — Mental and Physical Health · [standard](#) [dissociative](#) [ptsd](#)
[eating-disorders](#)

Dissociative disorders — disruptions of consciousness, memory, and identity, usually related to trauma or stress:

- **Dissociative amnesia:** memory loss for personal information — with or without **fugue** (travel away from home).
- **Dissociative identity disorder (DID):** two or more distinct identity states (“split” identity — not schizophrenia).

Trauma- and stressor-related disorders:

- **PTSD (posttraumatic stress disorder):** after trauma — hypervigilance, flashbacks, severe anxiety, nightmares, insomnia, emotional detachment, hostility.

Feeding and eating disorders:

- **Anorexia nervosa:** restriction of intake; intense fear of gaining weight; distorted body image.
- **Bulimia nervosa:** binge eating followed by compensatory purging.

Misconception: dissociative amnesia is psychological (usually trauma-linked), not neurological injury; and the anorexia/bulimia distinction is the PATTERN — restriction with distorted body image vs. binge-purge cycles (weight is not the defining criterion for bulimia).

Exam tip: PTSD questions hinge on the TRAUMA exposure plus the symptom cluster (intrusive re-experiencing, avoidance, hyperarousal); dissociative symptoms after a terrifying event may co-occur — keep the categories separate in the answer.

Personality Disorders

id: hp-personality-disorders · Unit 5 — Mental and Physical Health · [standard](#) [personality-disorders](#) [clusters](#)
[disorders](#)

Personality disorders: enduring, inflexible, and pervasive patterns of inner experience and behavior that deviate from cultural expectations, with onset by early adulthood, causing distress or impairment. Causes: biological, genetic, social, cultural, behavioral, and cognitive factors.

Cluster A — odd or eccentric:

- **Paranoid:** distrust and suspiciousness of others.
- **Schizoid:** detachment from social relationships; restricted emotional expression.
- **Schizotypal:** odd beliefs, magical thinking, and peculiar perceptions.

Cluster B — dramatic, emotional, or erratic:

- **Antisocial:** disregard for and violation of others' rights.
- **Histrionic:** excessive attention-seeking and emotionality.
- **Narcissistic:** grandiosity, need for admiration, lack of empathy.
- **Borderline:** instability of relationships, self-image, and emotions.

Cluster C — anxious or fearful:

- **Avoidant:** social inhibition from feelings of inadequacy.
- **Dependent:** excessive need to be cared for.
- **Obsessive-compulsive PD:** rigid perfectionism and control — DISTINCT from OCD (no obsessions/compulsions cycle).

Mnemonic: clusters as MAD — **A** = Awkward (odd), **B** = Bold/dramatic, **C** = Cautious (anxious).

Misconception: antisocial does not mean “shy” — it means acting AGAINST society (violating others' rights); and OCPD is ego-syntonic perfectionism (the person sees no problem), whereas OCD is ego-dystonic (the obsessions are unwanted and distressing).

Exam tip: the three-way discrimination the exam loves — antisocial PD (behavior against others' rights), narcissistic PD (grandiosity), borderline PD (relational and self-image instability) — match the scenario descriptors to the cluster pattern.

Psychotherapies

id: hp-psychotherapy · Unit 5 — Mental and Physical Health · core therapy cbt treatment

therapeutic-alliance

Research and trends: meta-analytic studies conclude psychotherapies are GENERALLY effective. **Therapeutic alliance** — the bond between therapist and client — is critical for success. **Cultural humility:** therapists' openness to clients' cultural backgrounds. **Evidence-based interventions:** treatment plans grounded in research. **Deinstitutionalization:** the late-20th-century release of massive numbers of patients from hospitals and asylums because of effective psychotropic medications — treatment is now decentralized and community-based, often combining medication and psychotherapy.

APA ethical principles for therapists: **nonmaleficence** (do no harm), **fidelity** (trustworthy relationships), **integrity**, and **respect for people's rights and dignity**.

Therapeutic approaches:

- **Psychodynamic therapy:** uncover the unconscious via **free association** (say whatever comes to mind) and **dream interpretation**.
- **Cognitive therapy:** **cognitive restructuring** (challenge and replace maladaptive thoughts); **fear hierarchies** (ranked anxiety triggers); target the **cognitive triad** — negative thoughts about the SELF, the WORLD, and the FUTURE.
- **Applied behavior analysis:** conditioning principles applied to disorders and developmental disabilities.
 - **Exposure therapies** — including **systematic desensitization** (relaxation paired with gradual exposure up a fear hierarchy).
 - **Aversion therapy:** pairing an undesirable behavior with an unpleasant stimulus.
 - **Token economies:** desirable behaviors earn tokens exchangeable for rewards.

- **Biofeedback:** electronic monitoring of body states (e.g., sympathetic arousal) so clients learn to regulate them.
- **Cognitive-behavioral therapy (CBT):** combines cognitive and behavioral techniques — **dialectical behavior therapy (DBT)** (emotional regulation, mindfulness; used for borderline personality disorder) and **rational-emotive behavior therapy (REBT)** (challenging irrational beliefs).
- **Humanistic/person-centered therapy (Rogers):** **active listening** (empathic, nonjudgmental reflection) and **unconditional positive regard**.
- **Group therapy:** multiple clients together — cost-effective, social support, universality — versus individual therapy’s privacy and focus.
- **Hypnosis:** effective for PAIN and ANXIETY reduction; NOT effective for recovering accurate memories or age regression.
- **(Self-help groups are taught: peer-led support groups.)**

Misconception: counterconditioning logic — systematic desensitization replaces fear with an INCOMPATIBLE response (relaxation); it is classical conditioning applied to treatment. And CBT is not “just positive thinking” — it tests and restructures specific maladaptive thoughts.

Exam tip: match the disorder to the therapy — specific phobia: systematic desensitization/exposure; depression: cognitive therapy (cognitive triad); borderline PD: DBT; schizophrenia: medication first (biomedical entry).

Biomedical Treatments

id: hp-biomedical-treatments · Unit 5 — Mental and Physical Health · **standard** medication ect tms treatment

Biological (biomedical) interventions change the body to treat psychological disorders:

- **Antidepressants: SSRIs** — increase serotonin availability by blocking its reuptake — for depression and anxiety.
- **Antianxiety drugs: benzodiazepines** — GABA agonists (they calm neural firing).
- **Lithium:** the mood stabilizer for bipolar disorder.
- **Antipsychotics:** block dopamine — for schizophrenia.
- **Side effects:** for example, **tardive dyskinesia** — an involuntary movement disorder from long-term antipsychotic use (an effect of dopamine regulation).
- **Psychosurgery:** lesioning; the **lobotomy** (severing frontal-lobe connections) — historically used, now virtually abandoned.
- **TMS (transcranial magnetic stimulation):** magnetic pulses stimulating specific brain regions (e.g., for depression).
- **ECT (electroconvulsive therapy):** controlled, electrically induced seizures — effective for SEVERE DEPRESSION when other treatments fail.

Misconception: ECT today is done under anesthesia with muscle relaxants — it is a legitimate, evidence-based treatment for treatment-resistant depression, not the frightening movie version; and SSRIs take weeks to help, not hours.

Exam tip: pair drug class with neurotransmitter mechanism — SSRIs block serotonin reuptake; benzodiazepines are GABA agonists; antipsychotics block dopamine — and lithium is the ONLY mood stabilizer listed. ECT/TMS/psychosurgery are reserved for severe or treatment-resistant cases.